

Validation of converted leg6dof9musc model

The converted MJC model has been tested under the three categories below and with the accuracy of:

- Step 1: XML Conversion Validation

Check multi-body forward kinematics (using endpoints), approximation of custom/coupling joints & conditional/moving path points

Mean error: 0.0471 cm; std: 0.0339 cm

- Step 2: Muscle Kinematics Validation

Check muscle moment arms as indication how muscle wrap over joints

Mean error: 1.1152 cm; std: 1.2094 cm

- Step 3: Muscle Kinetic Validation

Check muscle force-length relationship as indication of how similar of them in generating forces

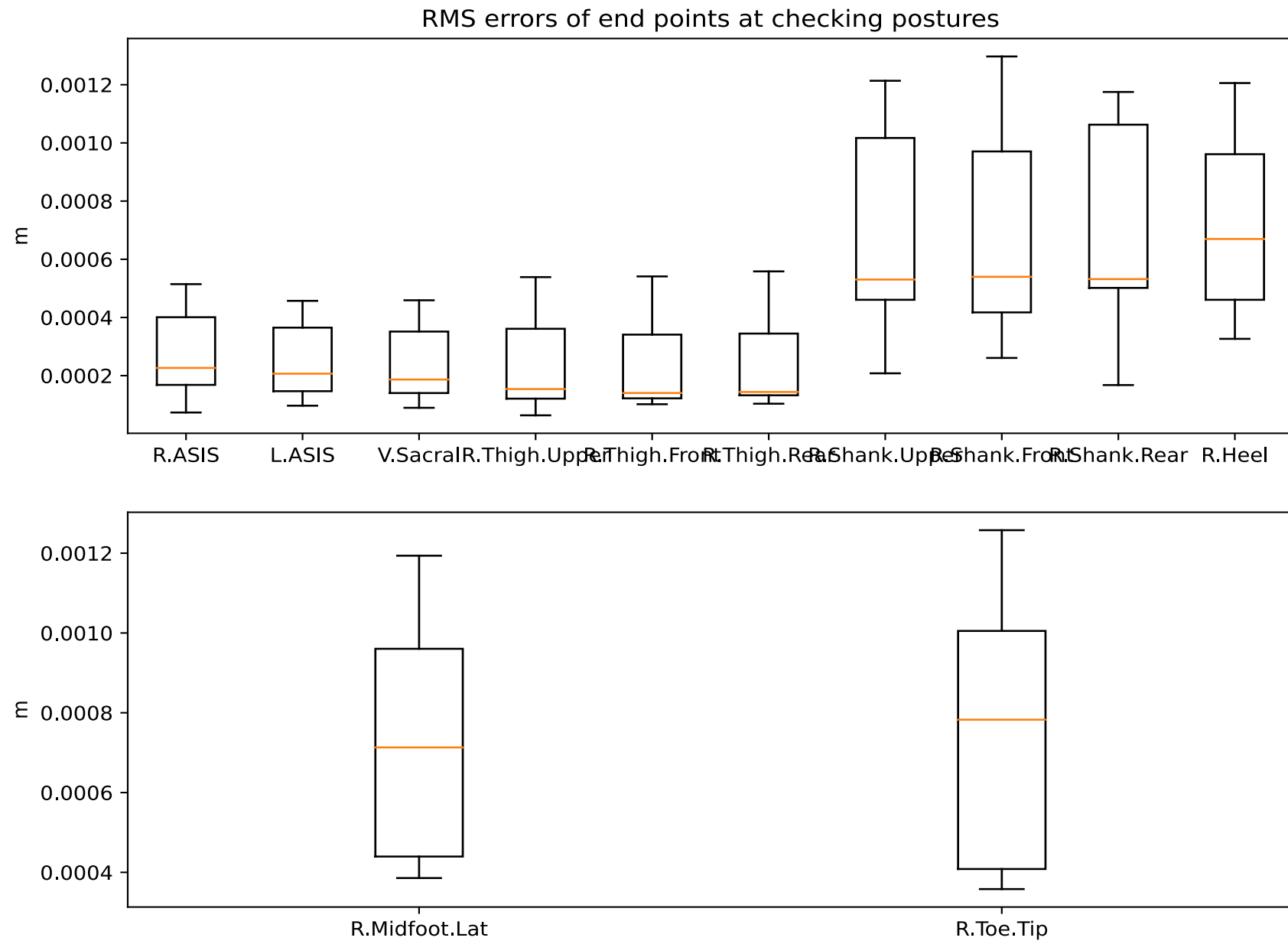
Mean error: 0.0547 Fmax; std: 0.0478 Fmax

Step 1: xml Conversion Validation

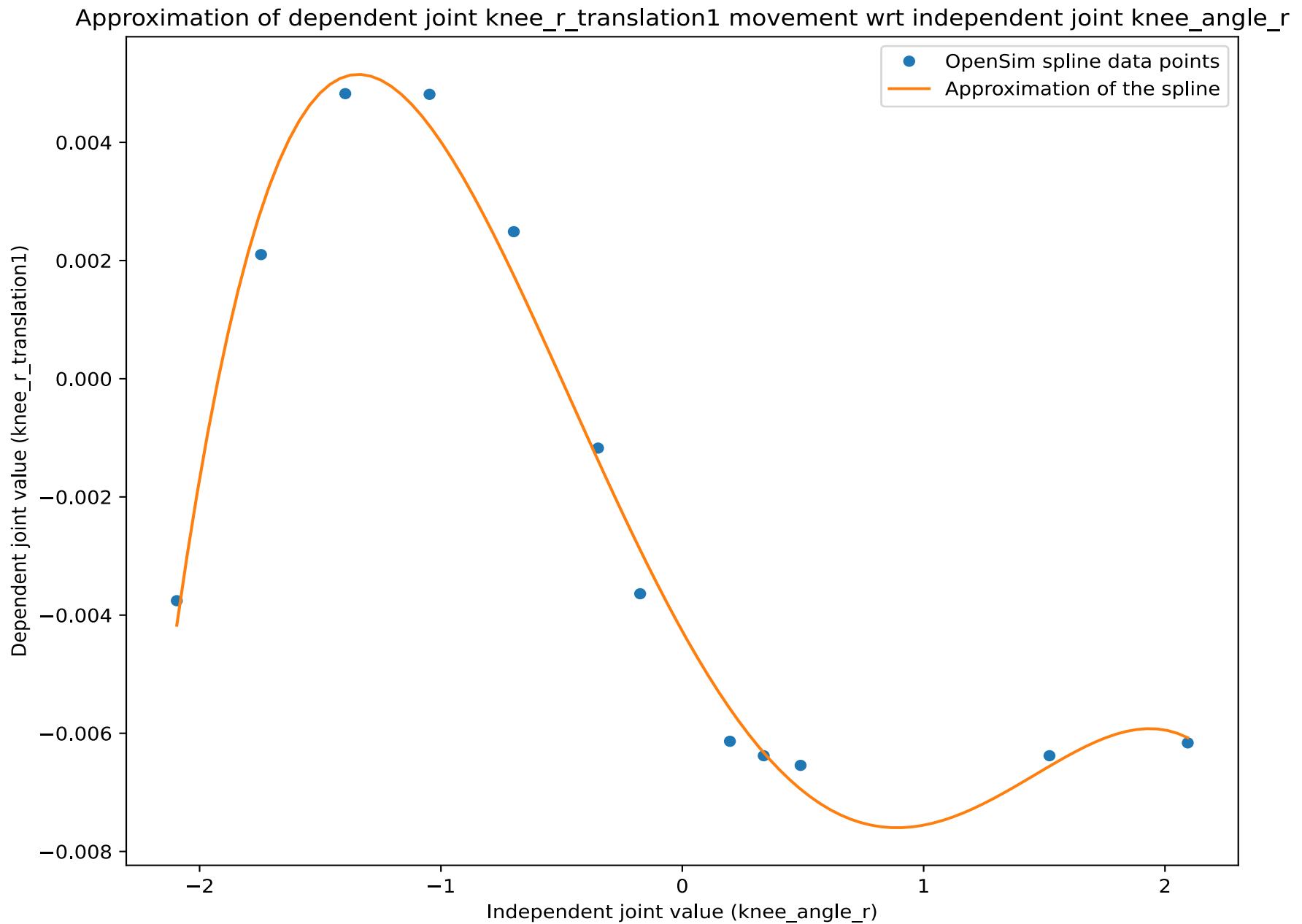
Randomly pose the model with 10 configurations within the joint limits. In each posture, the endpoints(markers) global locations of Osim and Mjc models are extracted and compared. Box plot of their mean-std errors are plotted together. Individual endpoint differences of these 10 postures are also plotted in the VLT folder, but not included inside this report.

Besides the endpoint check, the approximation of customer joints, coupling joints, conditional/moving path points are plotted and attached. In these plots, blue dots/lines represent their setup in the OpenSim model. Yellow dots/lines represent the approximations in the MuJoCo model.

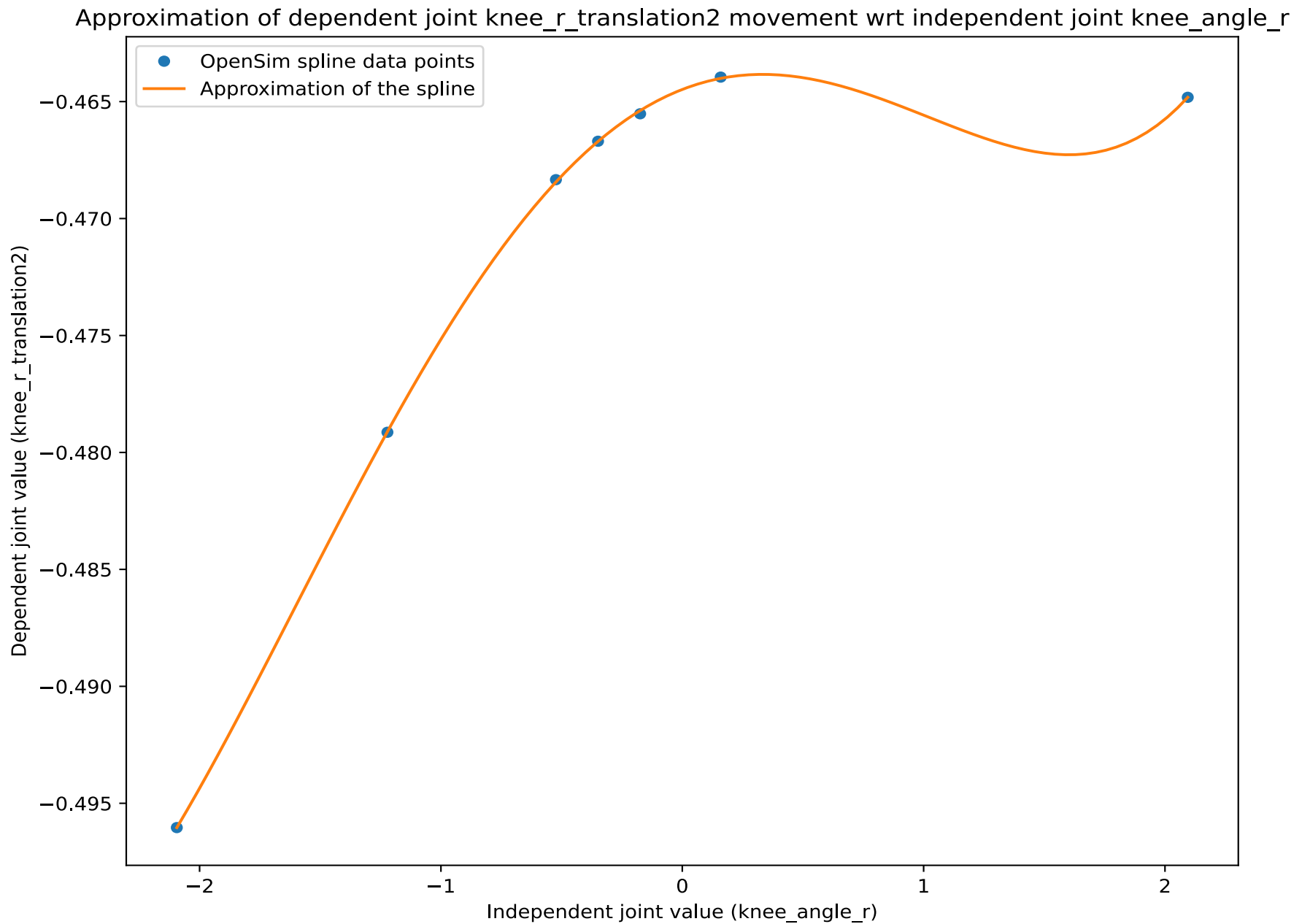
End points



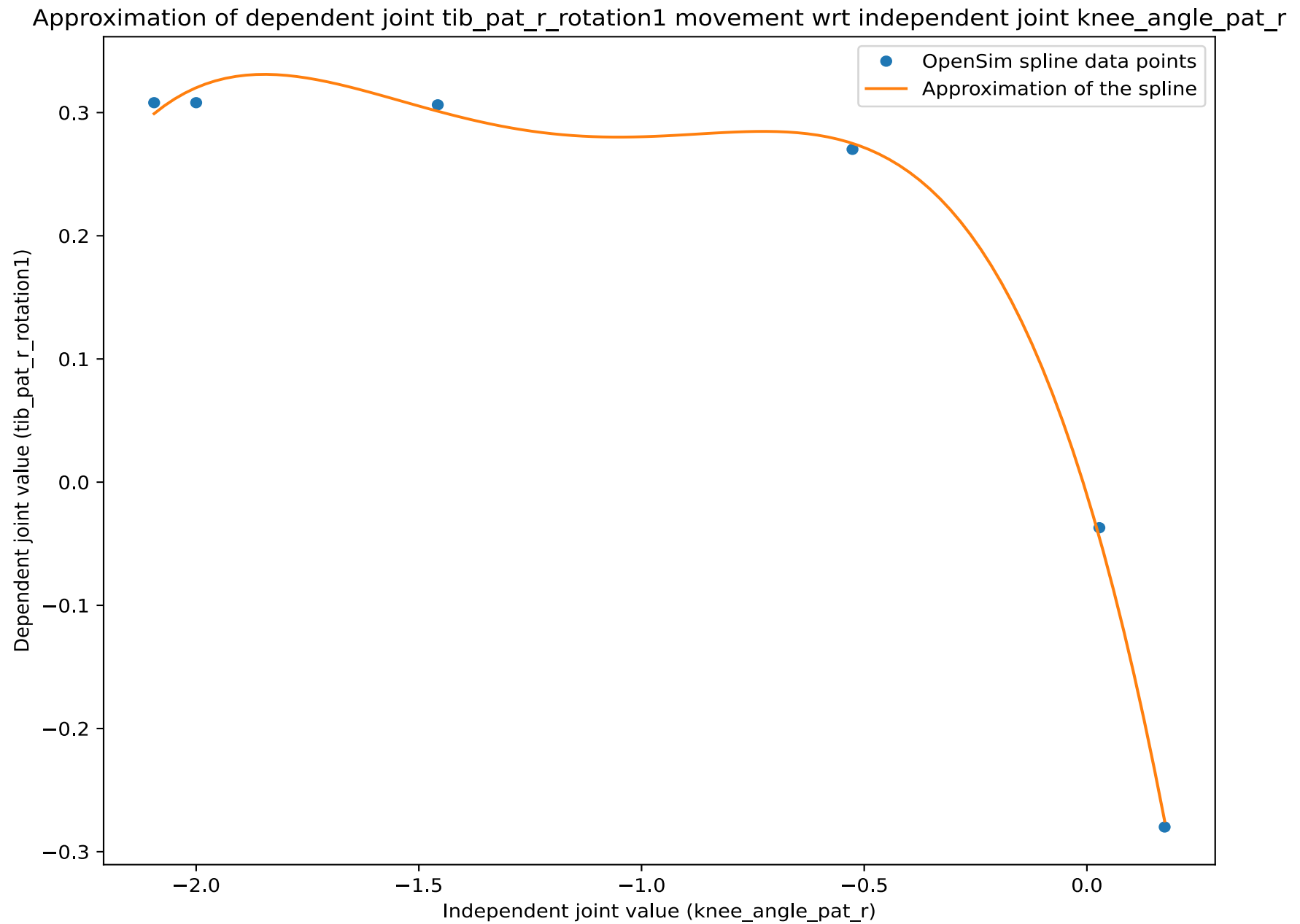
Approximation of custom joints



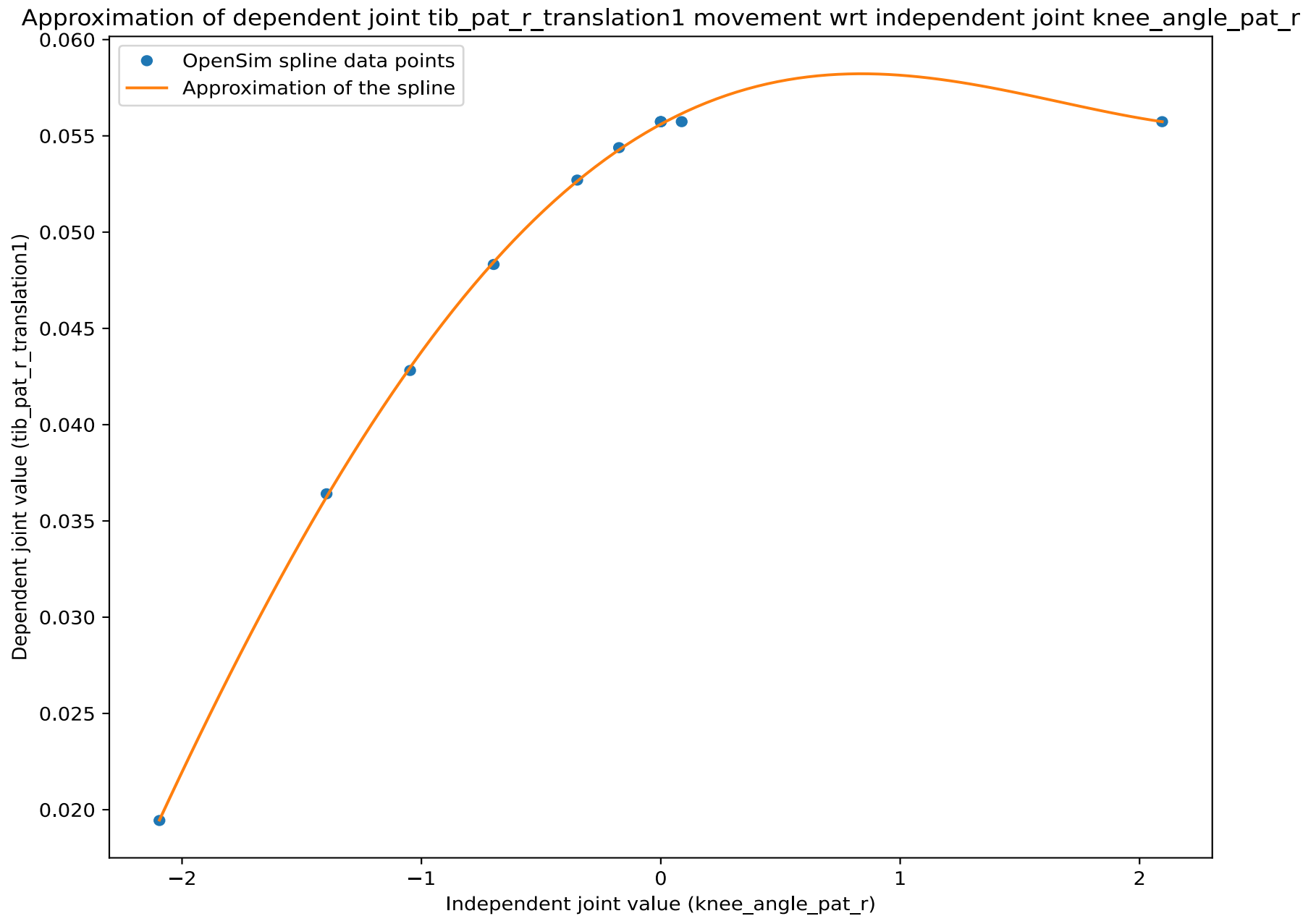
Approximation of custom joints



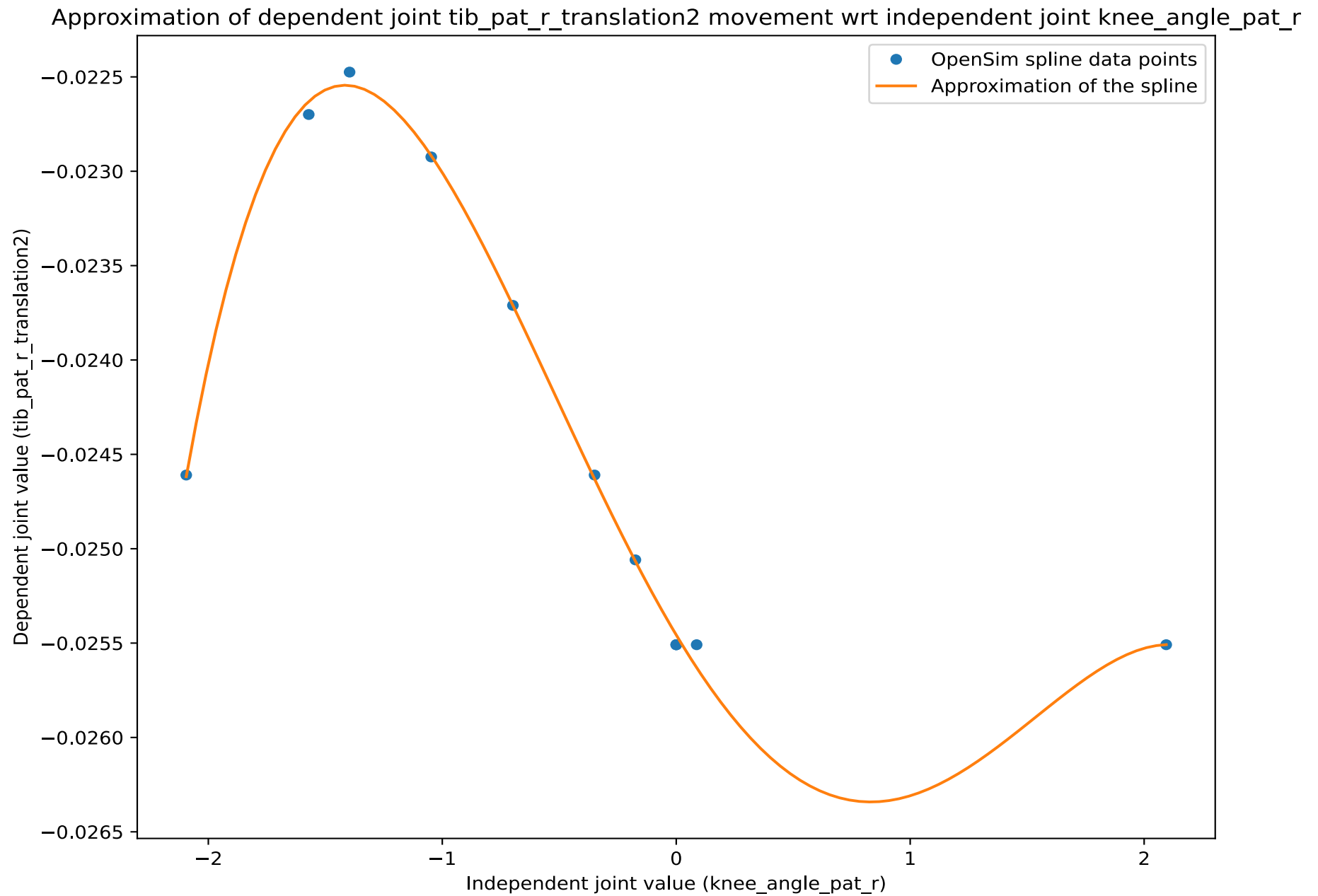
Approximation of custom joints



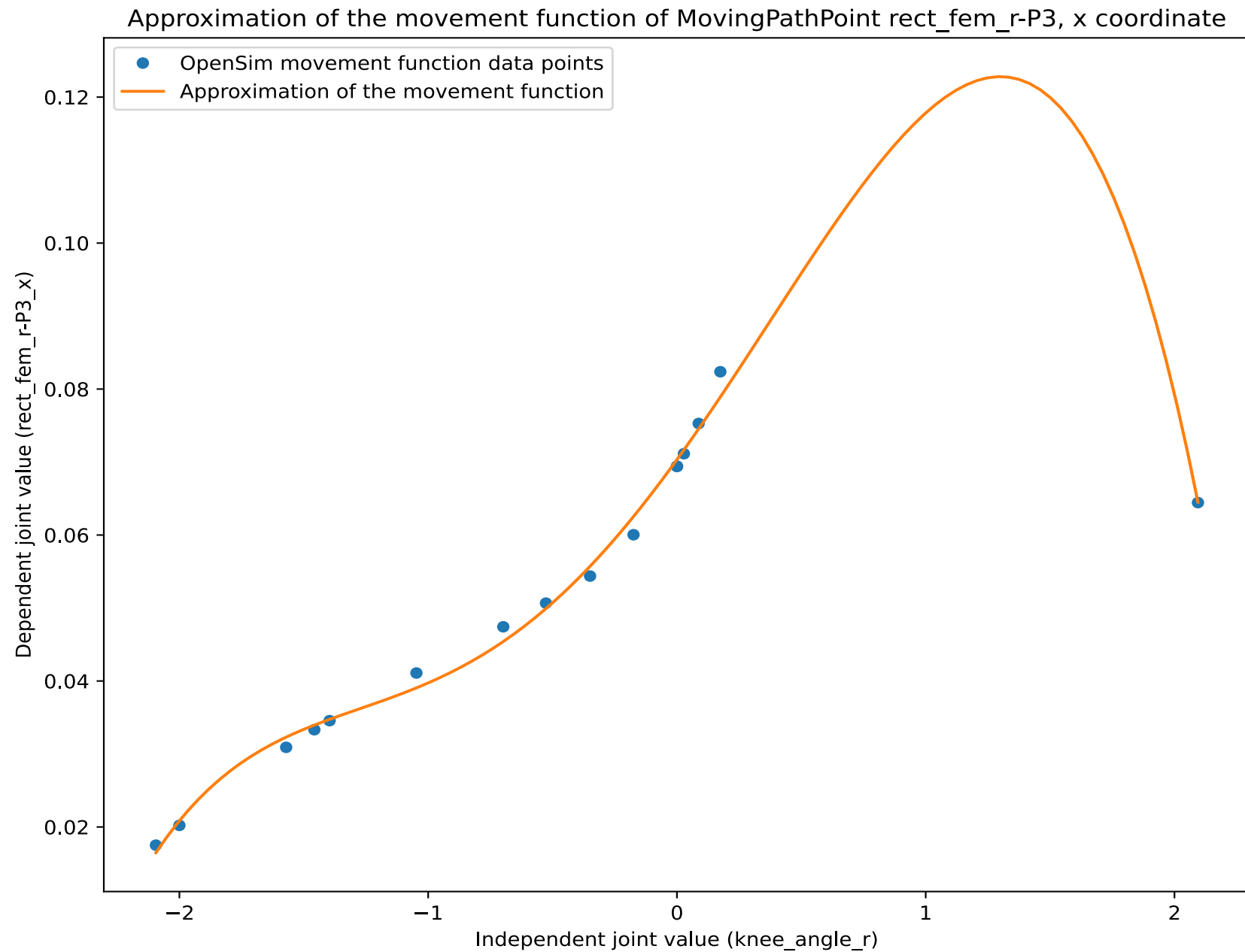
Approximation of custom joints



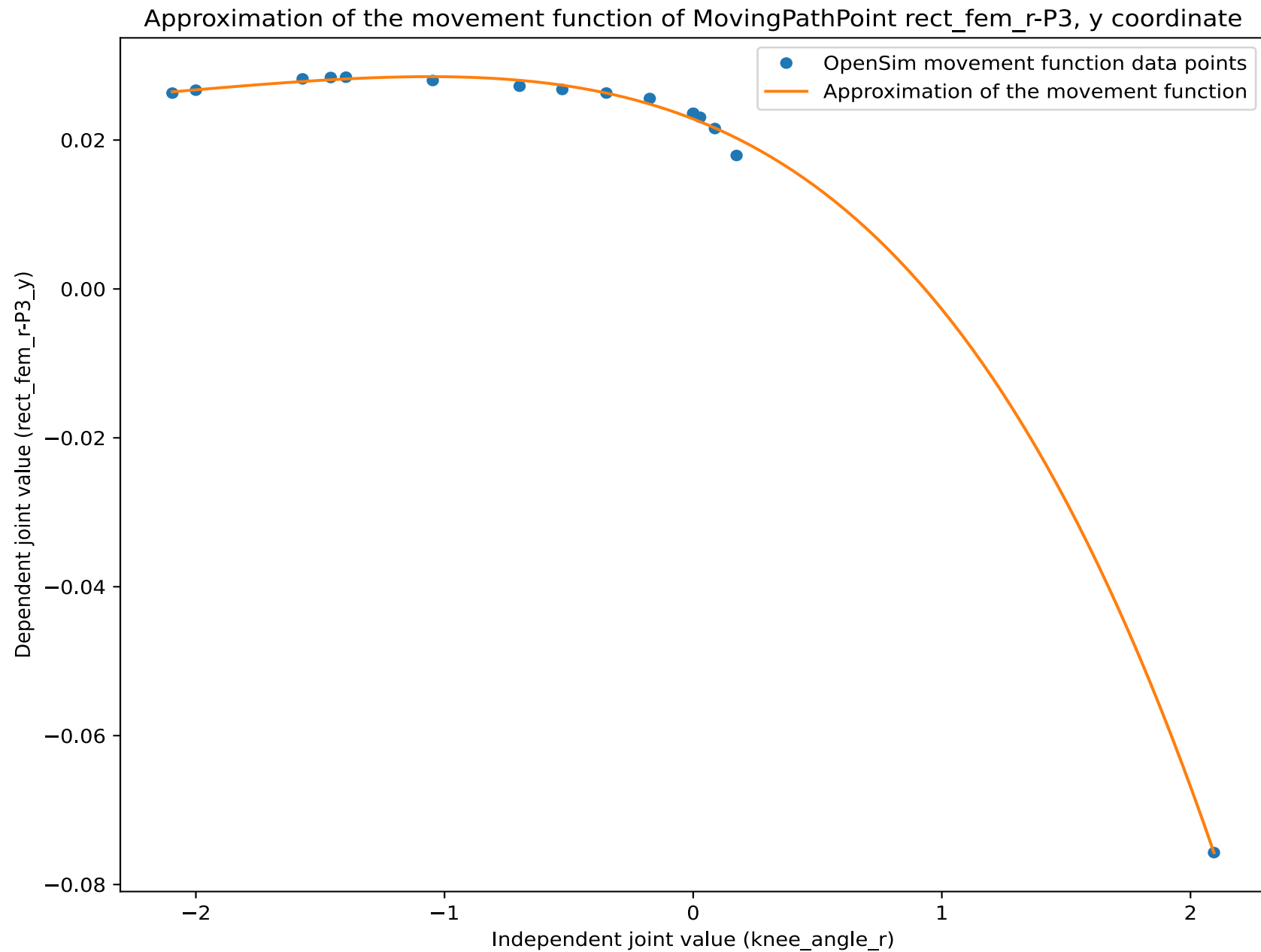
Approximation of custom joints



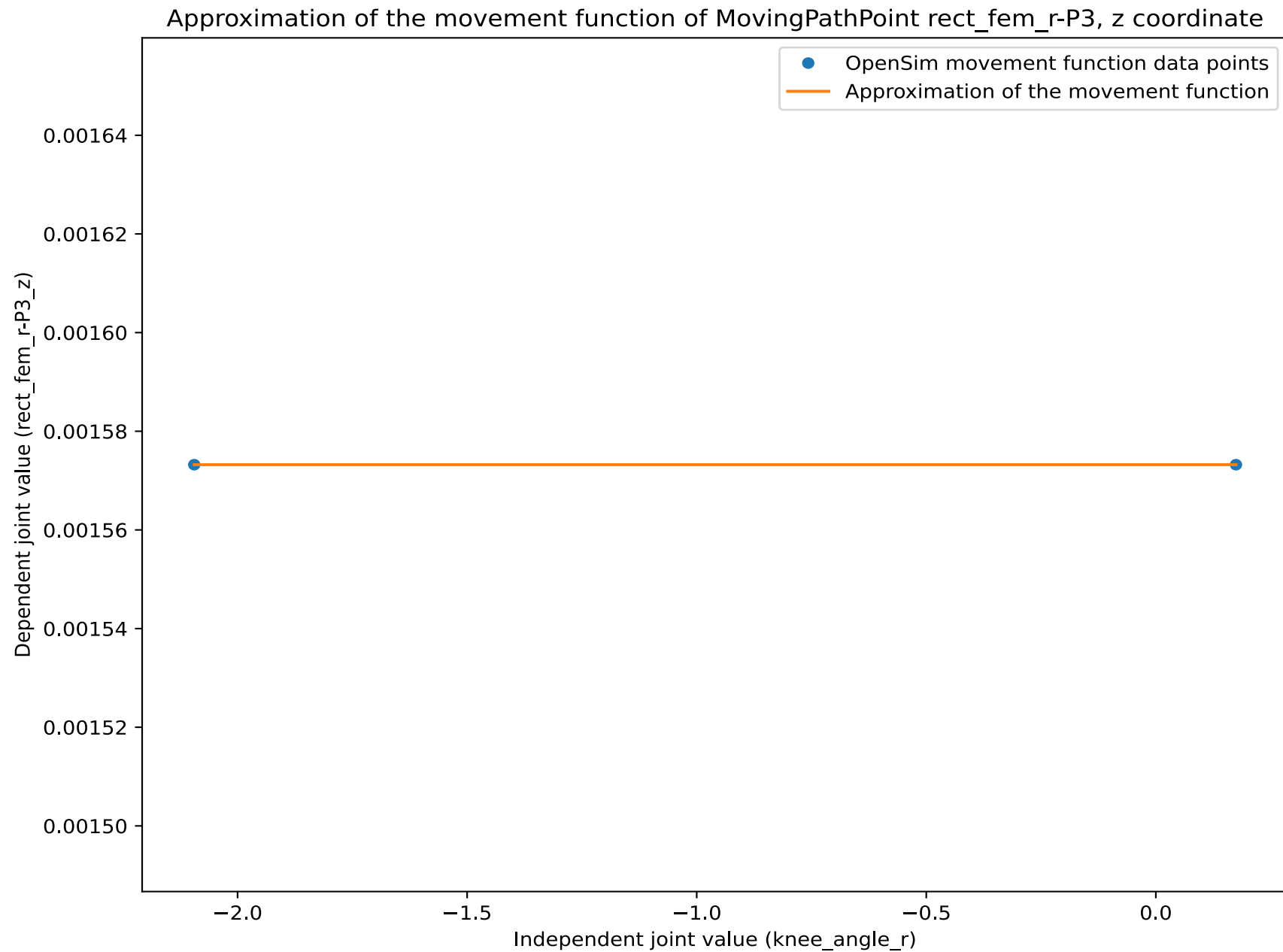
Approximation of moving path points



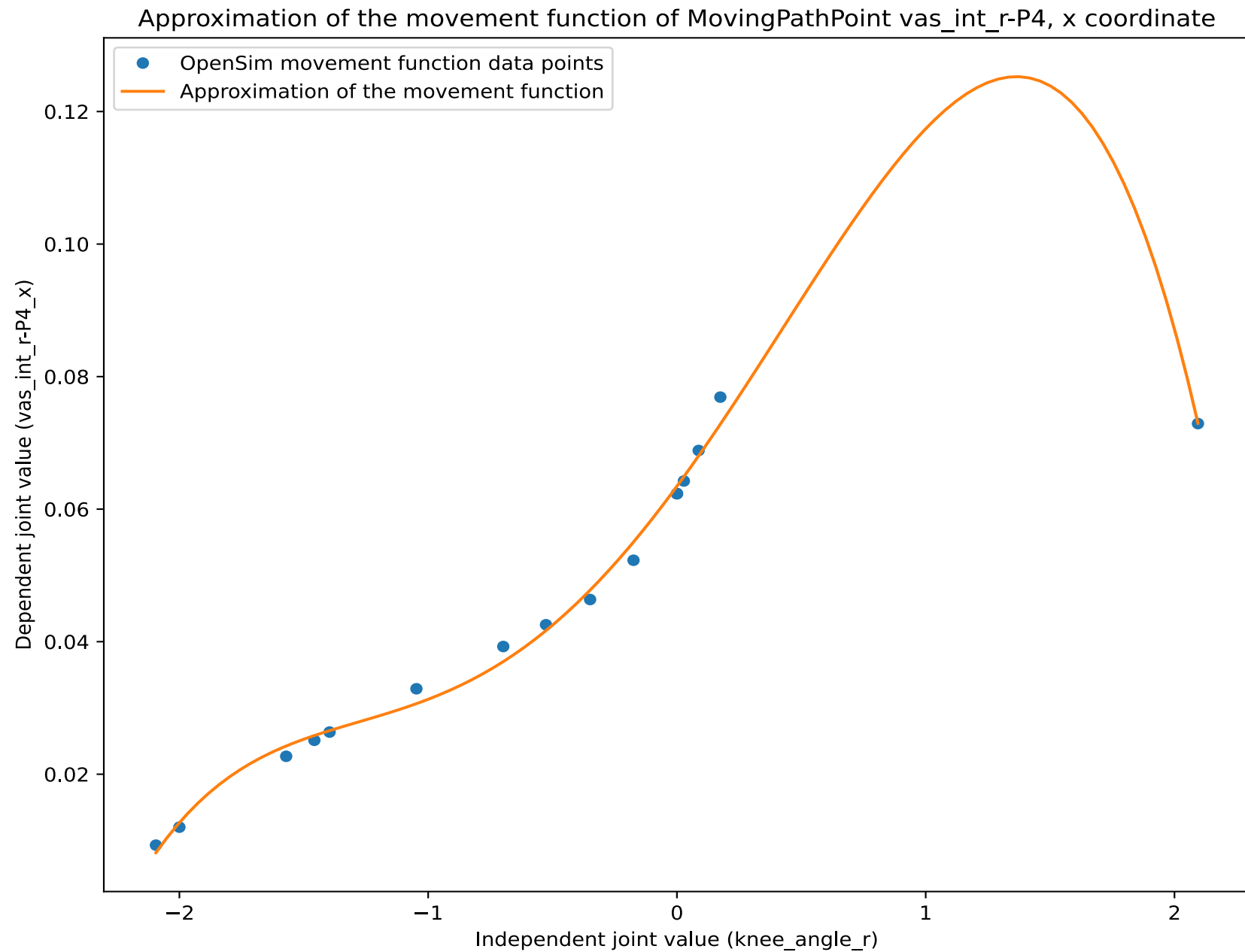
Approximation of moving path points



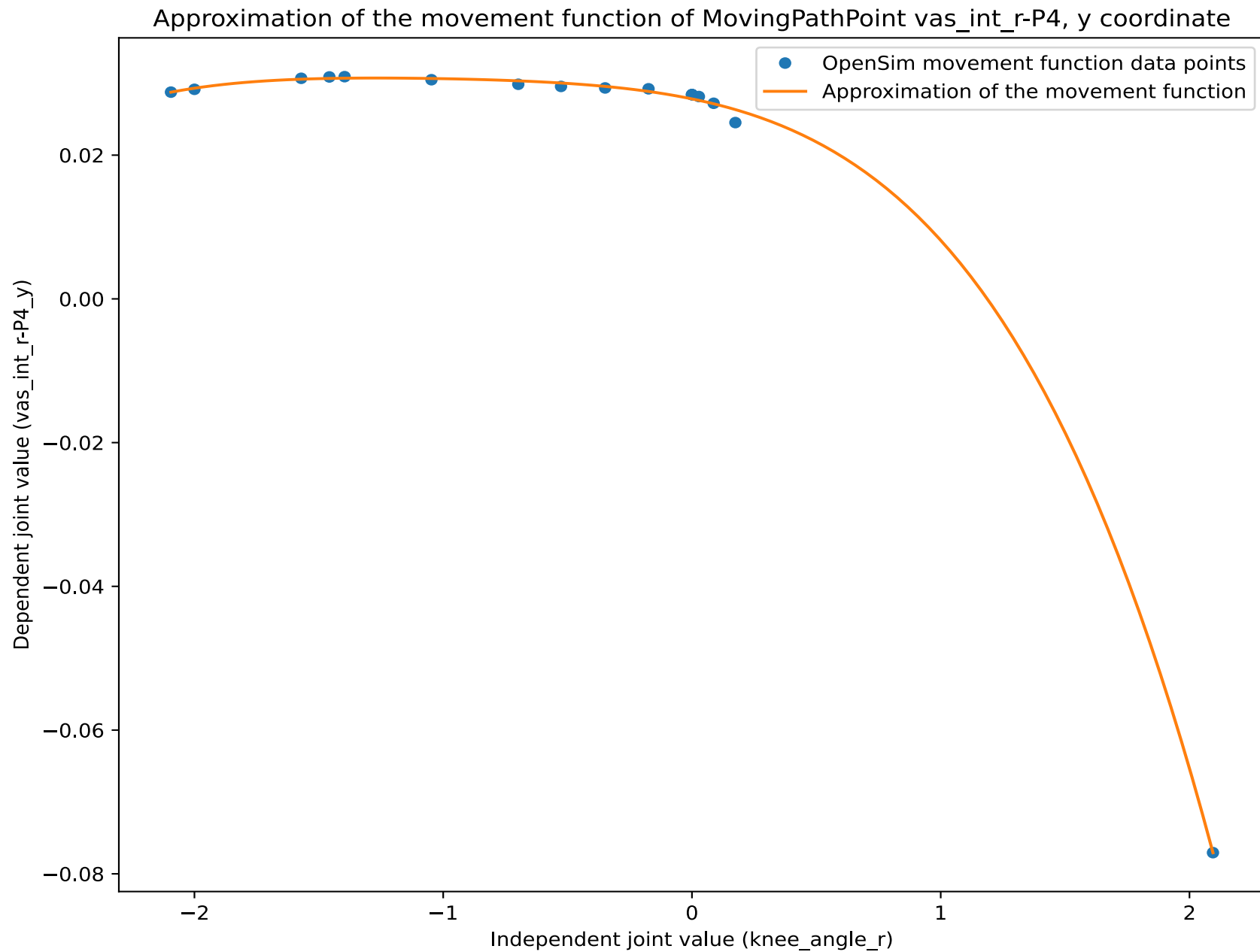
Approximation of moving path points



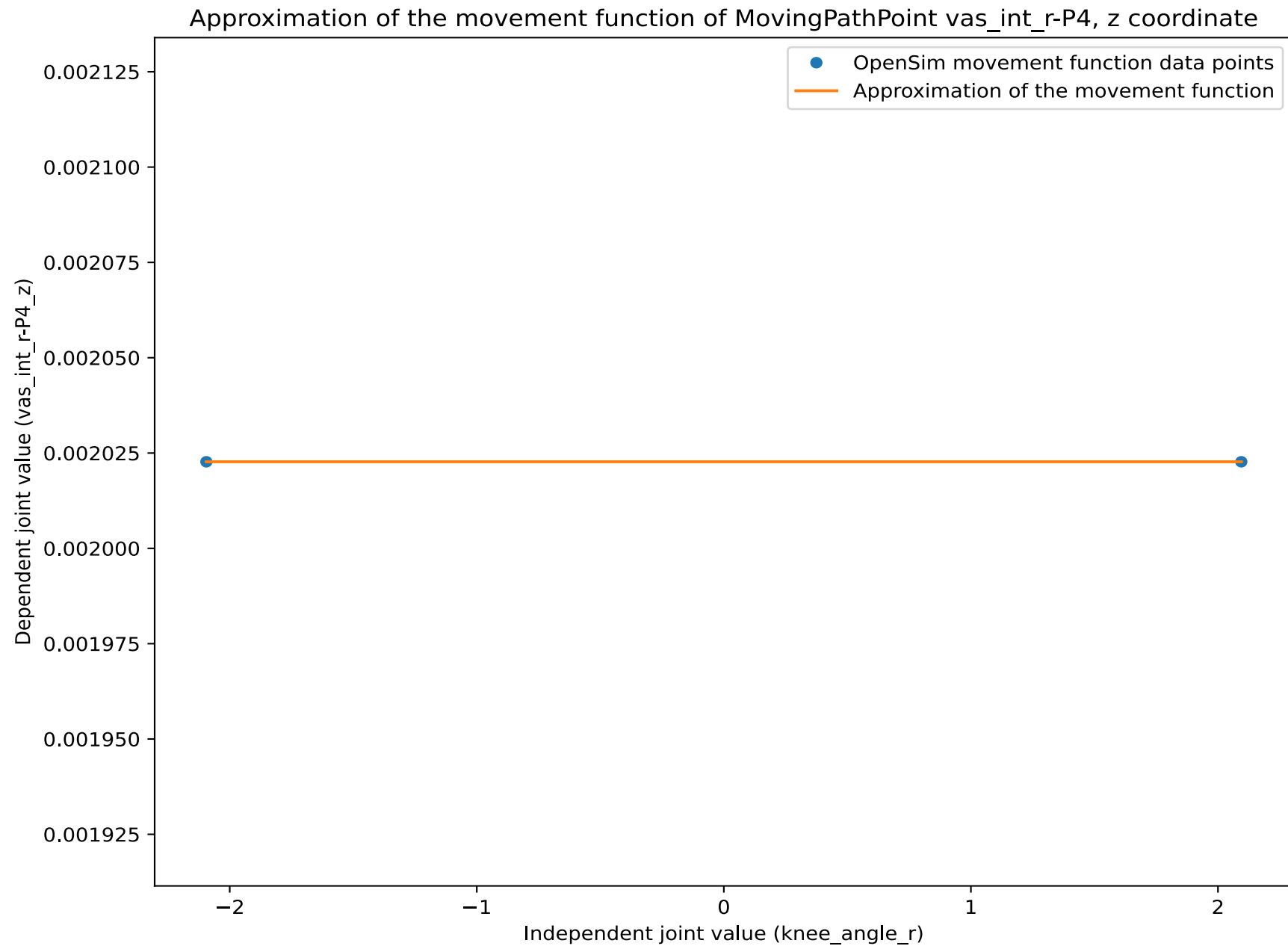
Approximation of moving path points



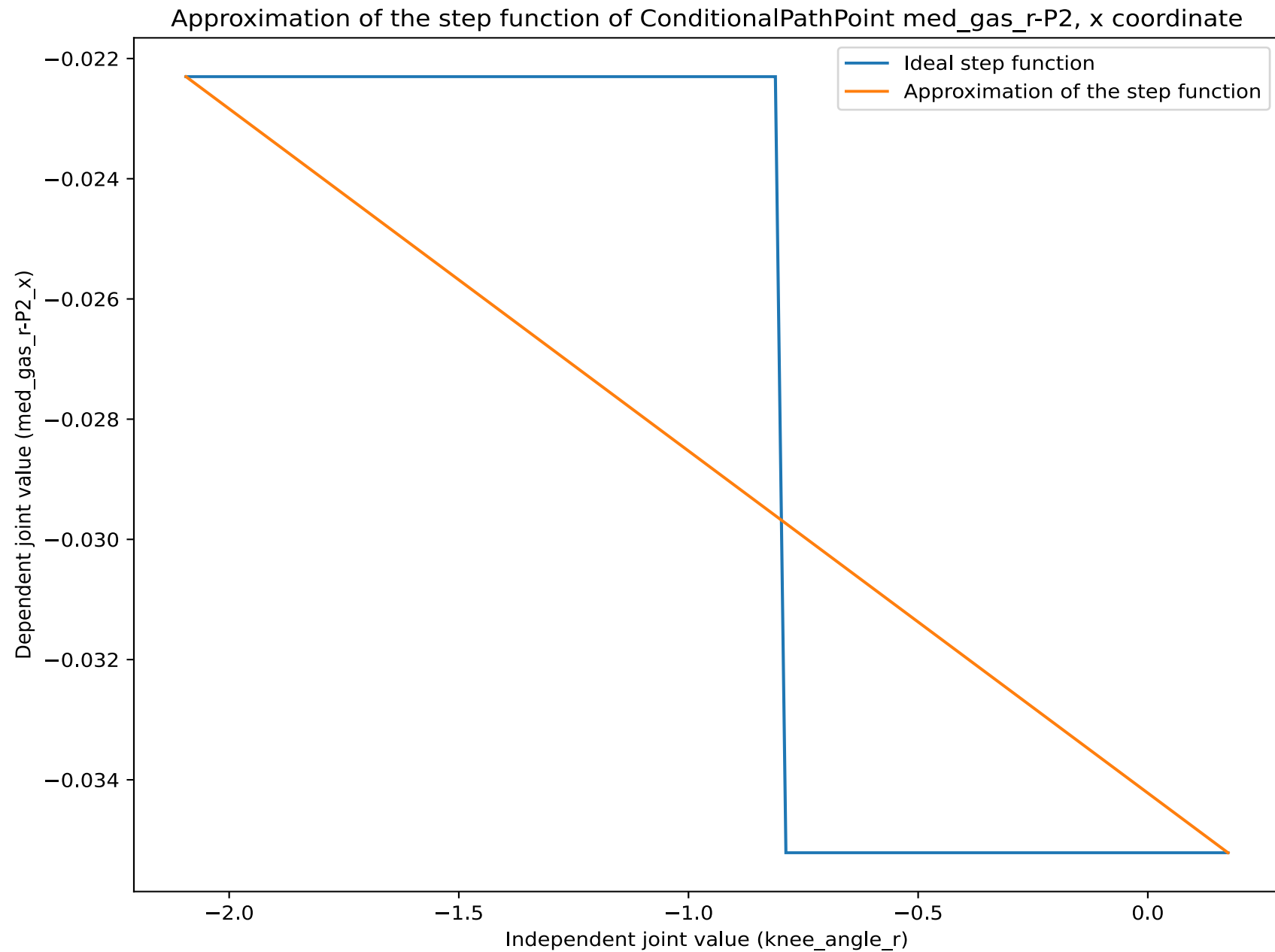
Approximation of moving path points



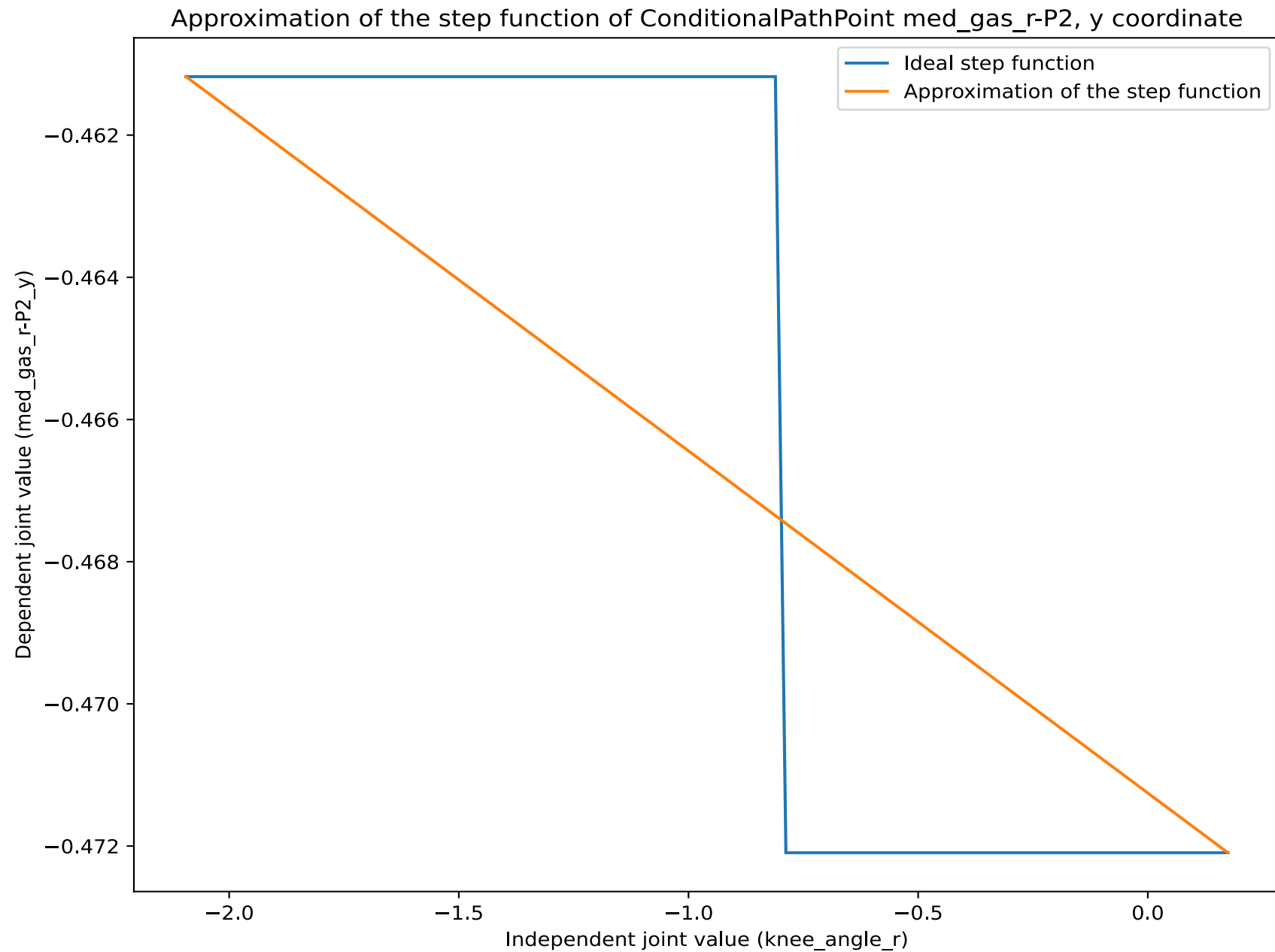
Approximation of moving path points



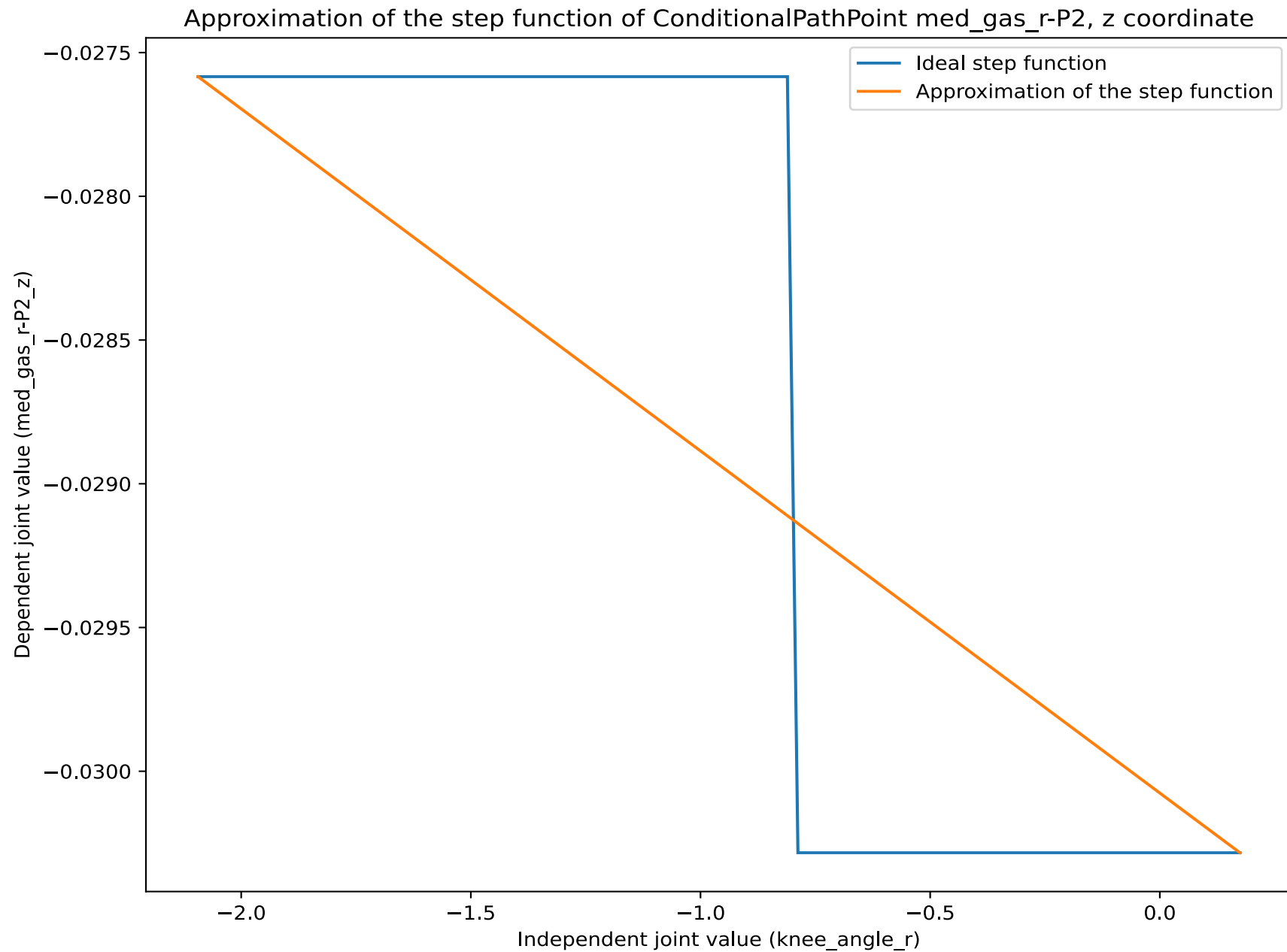
Approximation of conditional path points



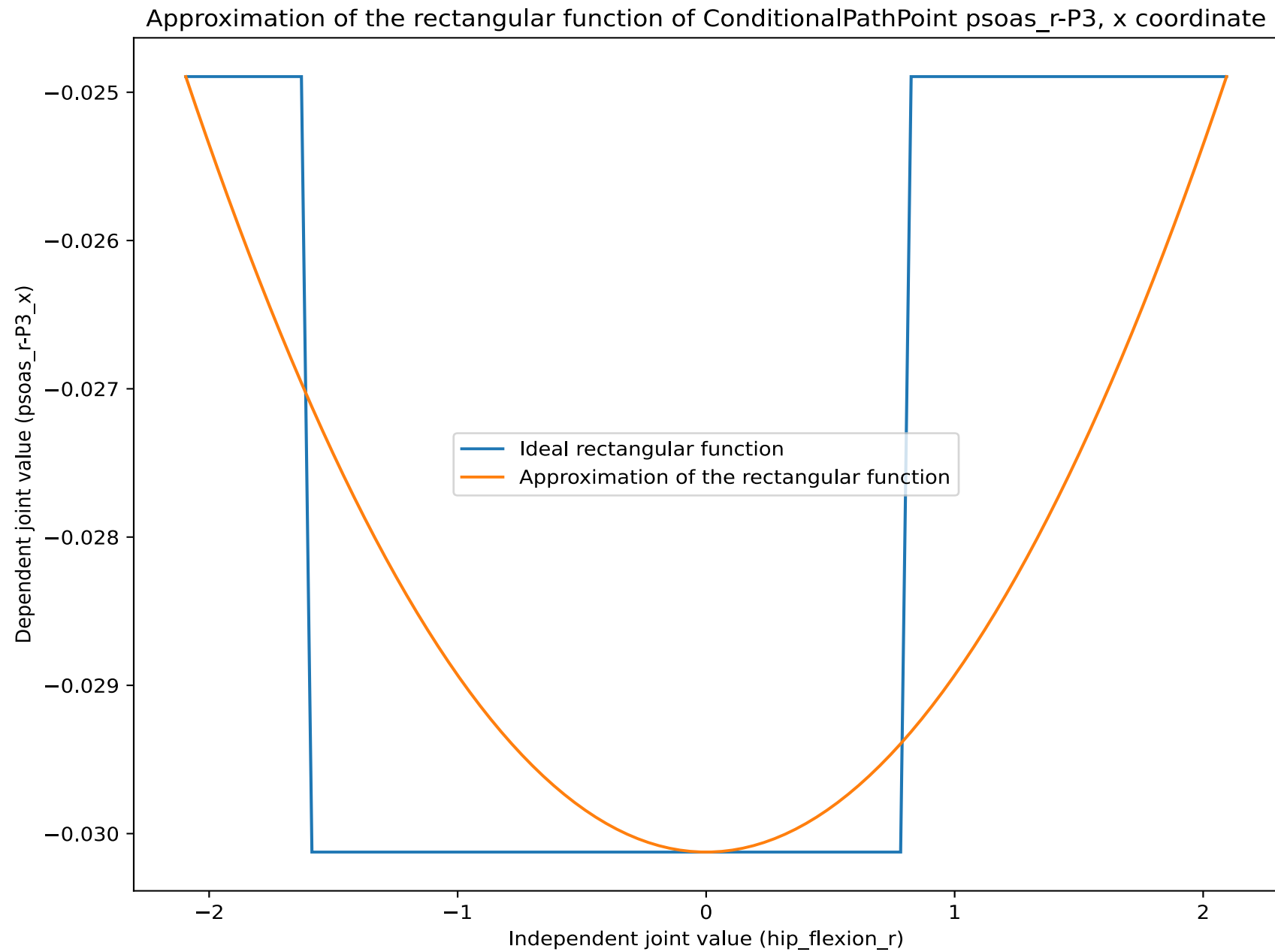
Approximation of conditional path points



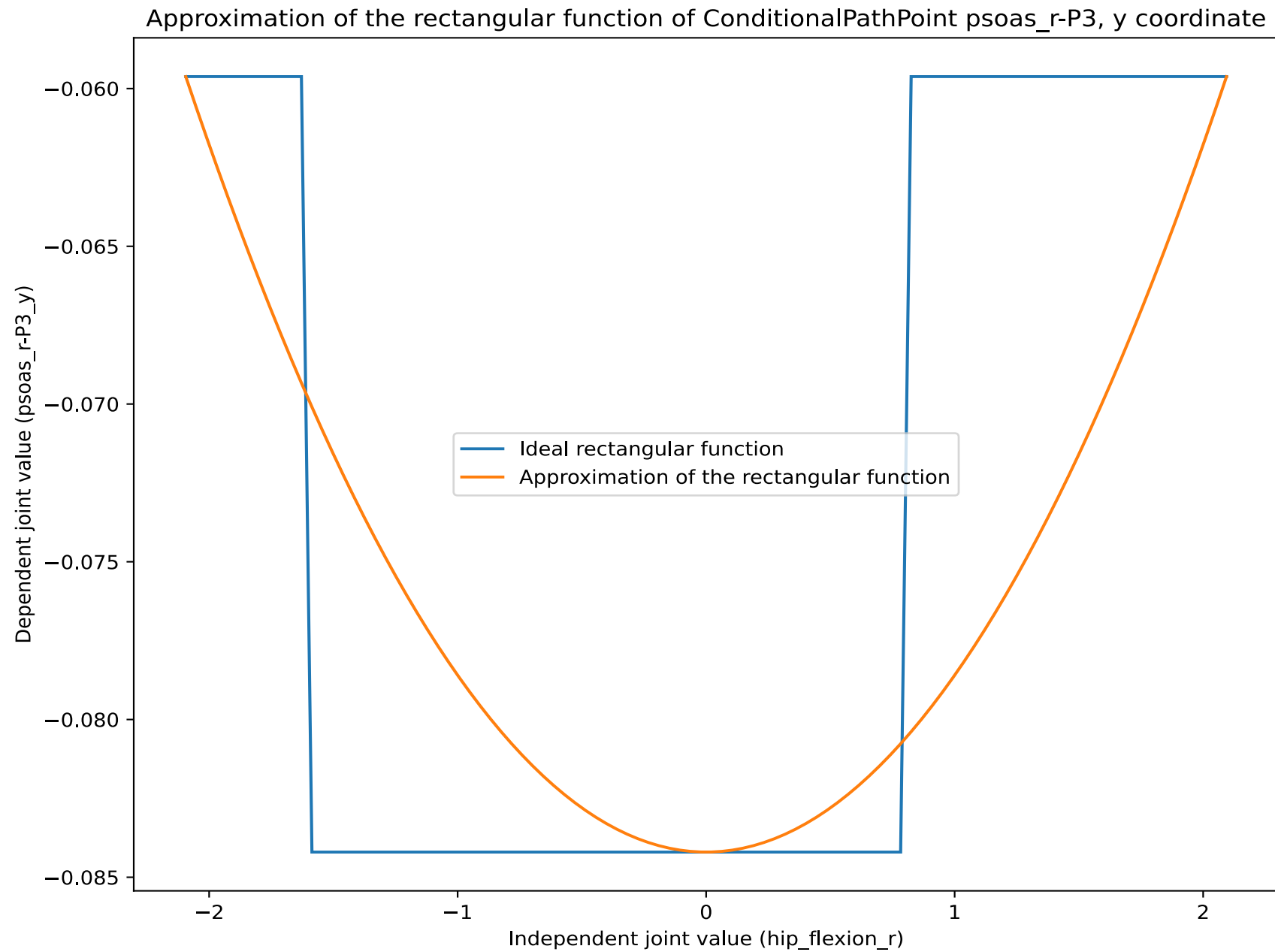
Approximation of conditional path points



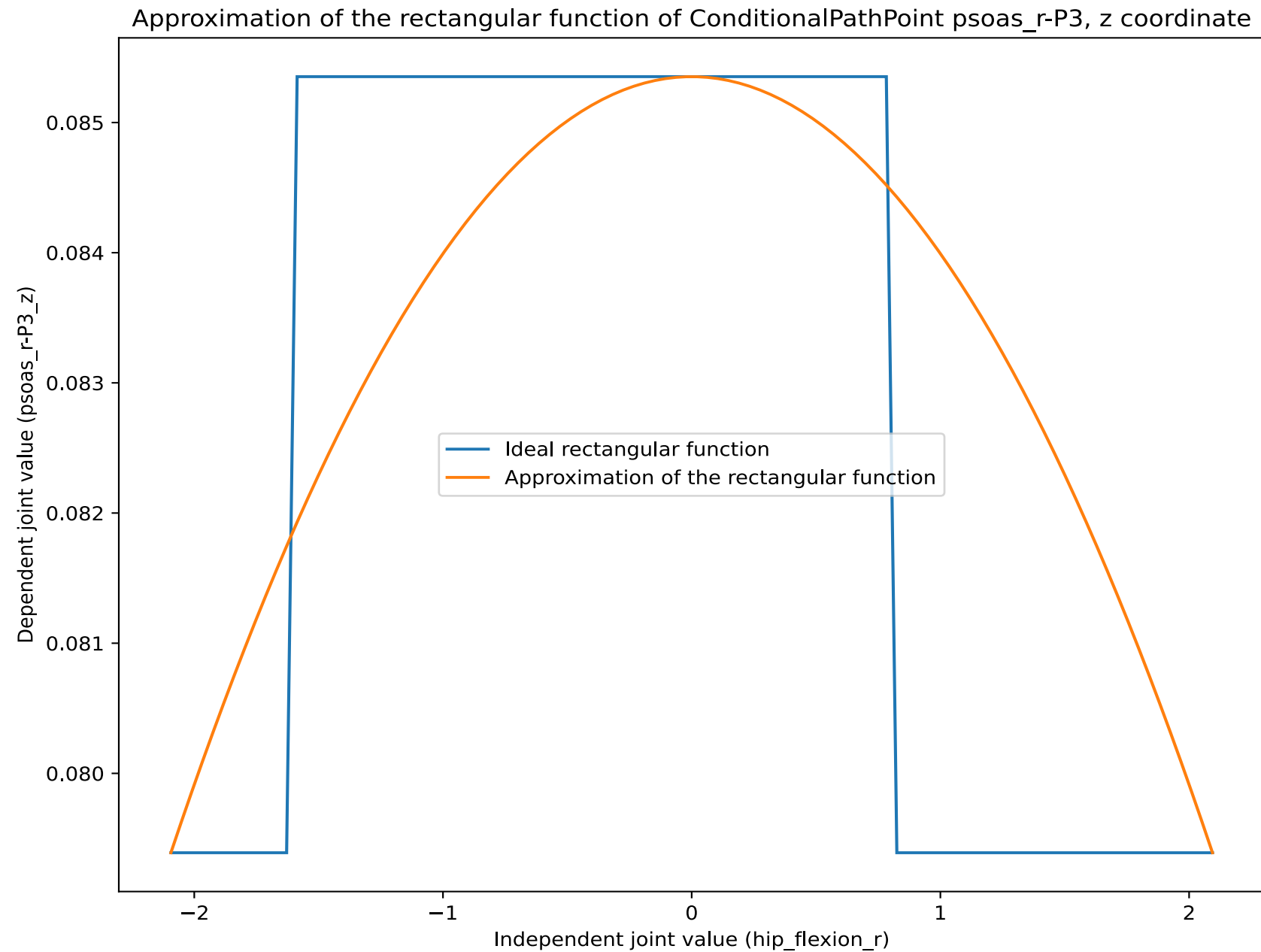
Approximation of conditional path points



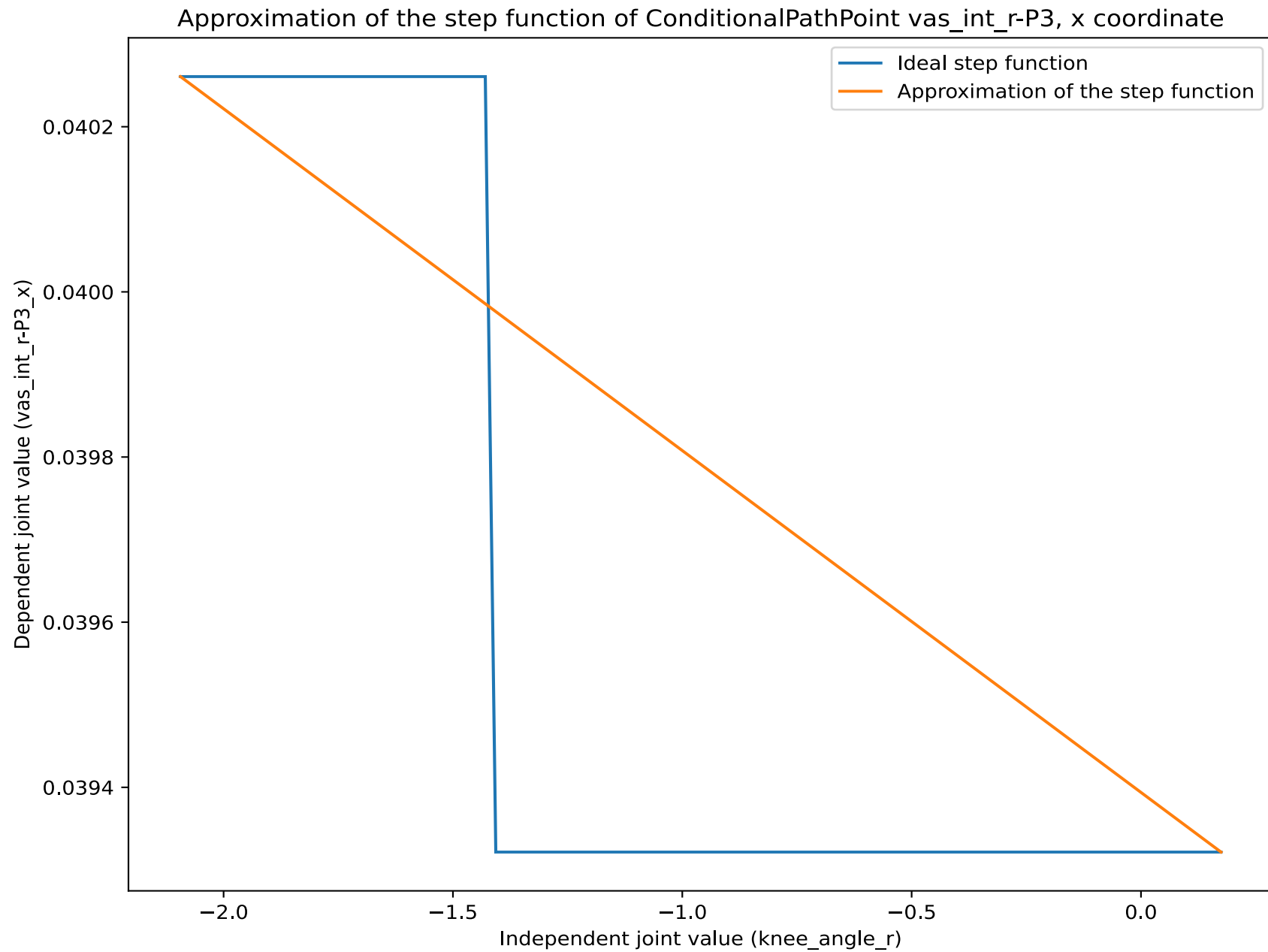
Approximation of conditional path points



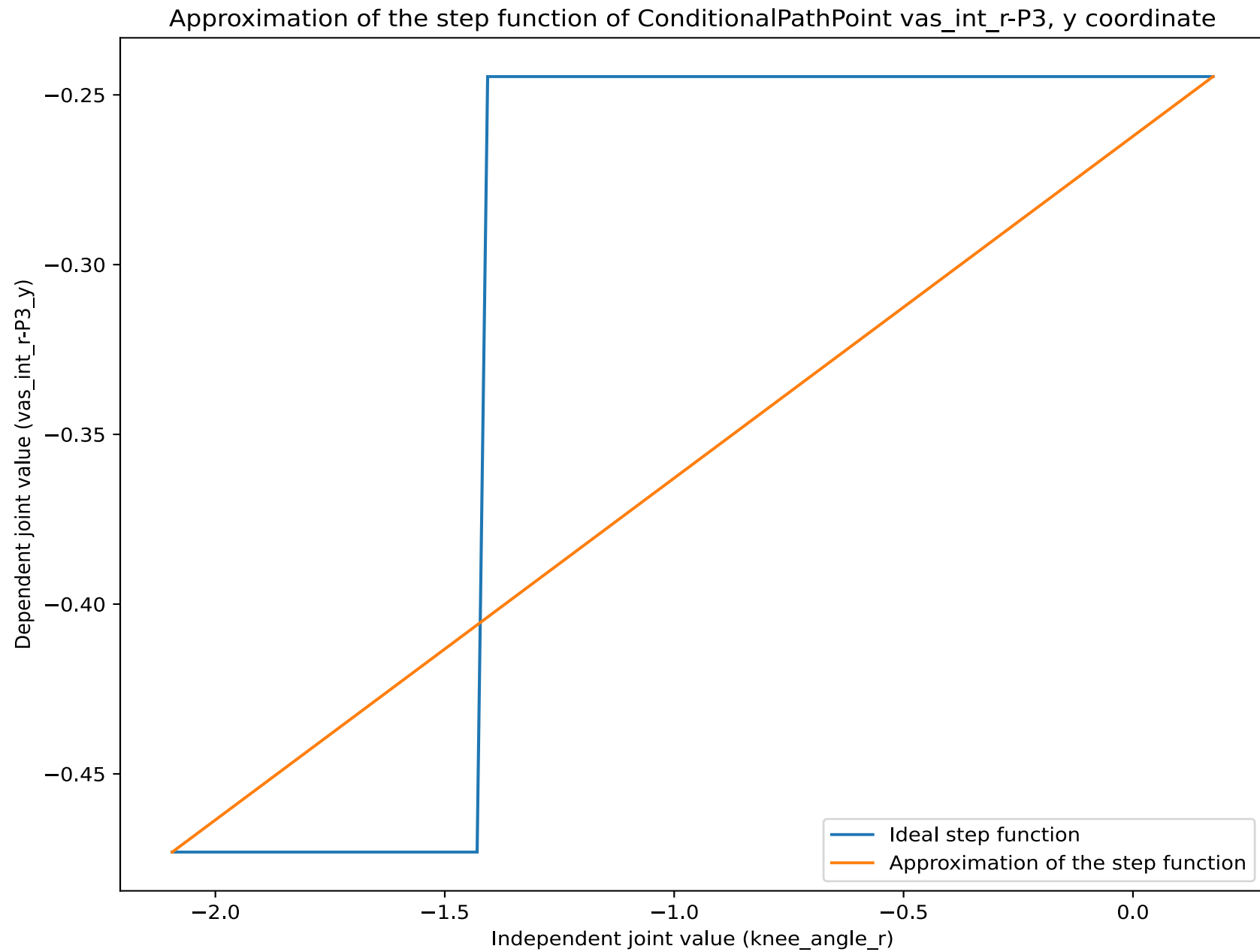
Approximation of conditional path points



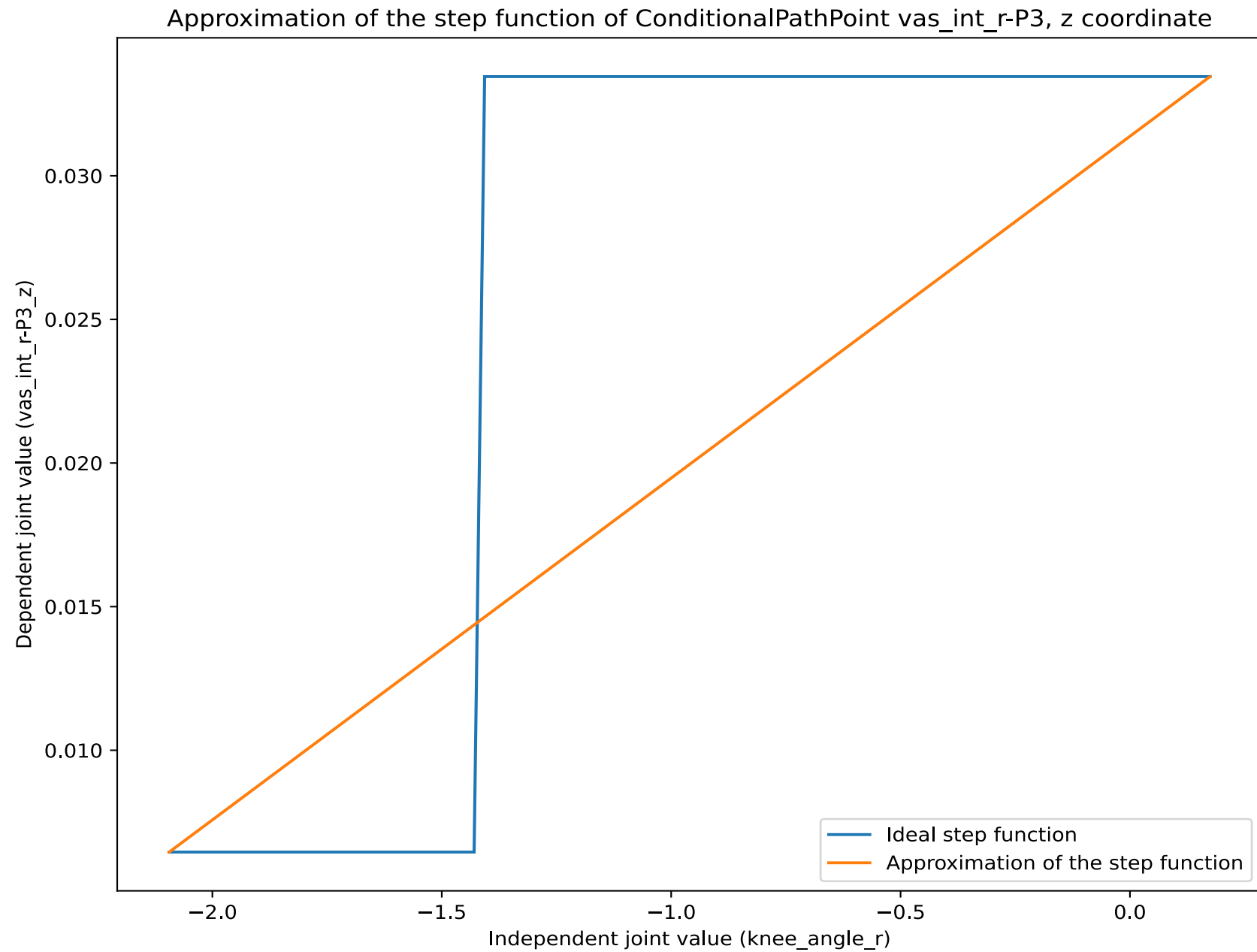
Approximation of conditional path points



Approximation of conditional path points



Approximation of conditional path points



Step 2: Muscle Kinematics Validation

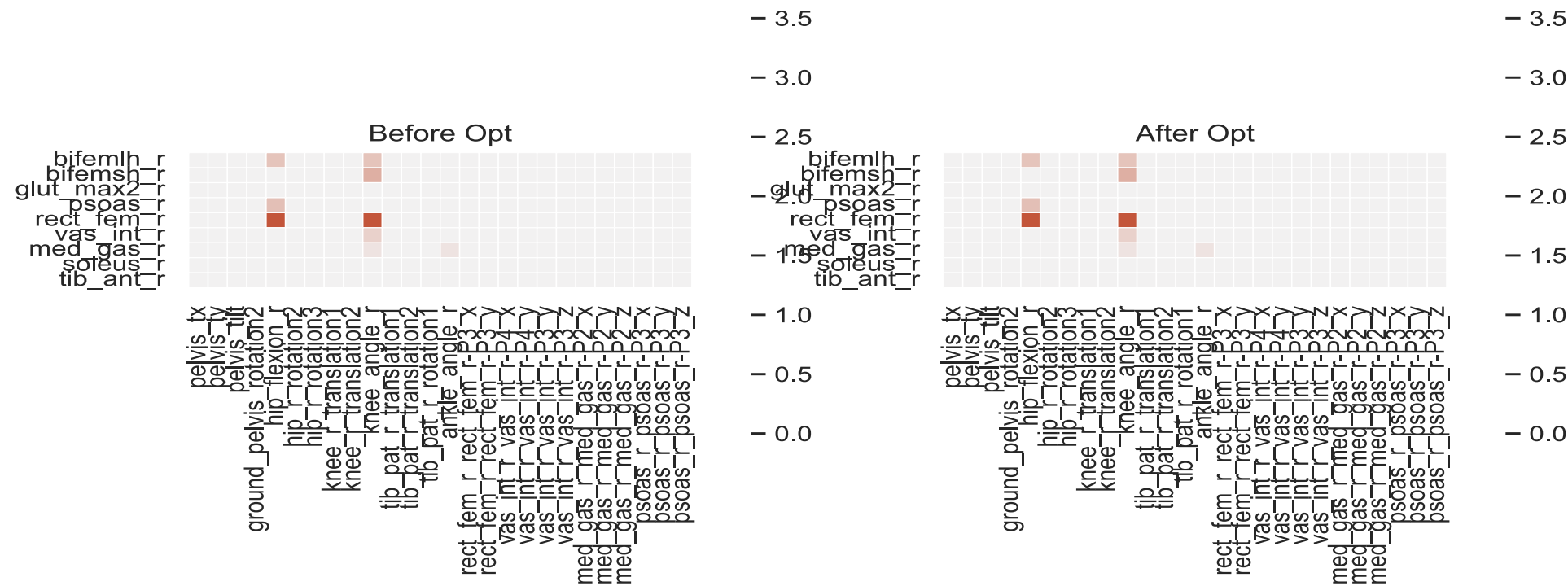
Moment arm of each muscle at each joint are compared between Osim and converted MJC model. A overall heatmap is included to indicate the overall moment arm errors before and after optimization. Then detail moment arm curves are plotted for comparison. For the muscles that wrap over multiple joints, moment arms with respect to one joint maybe affected by several other joints. In this case, several mesh points were check of these affecting joints, when plotting moment arms at one joint. This is why there are multiple lines (with different grey levels) plotted for one muscle on one joint.

How to interpret the plot:

Global title indicate the muscle and joints that affecting the moment arms in the plots. X axis indicate the joint that moment arms were extracted. Grey level of the lines indicate the mesh postures of other relevant joints (in the global tile, but not the x axis)

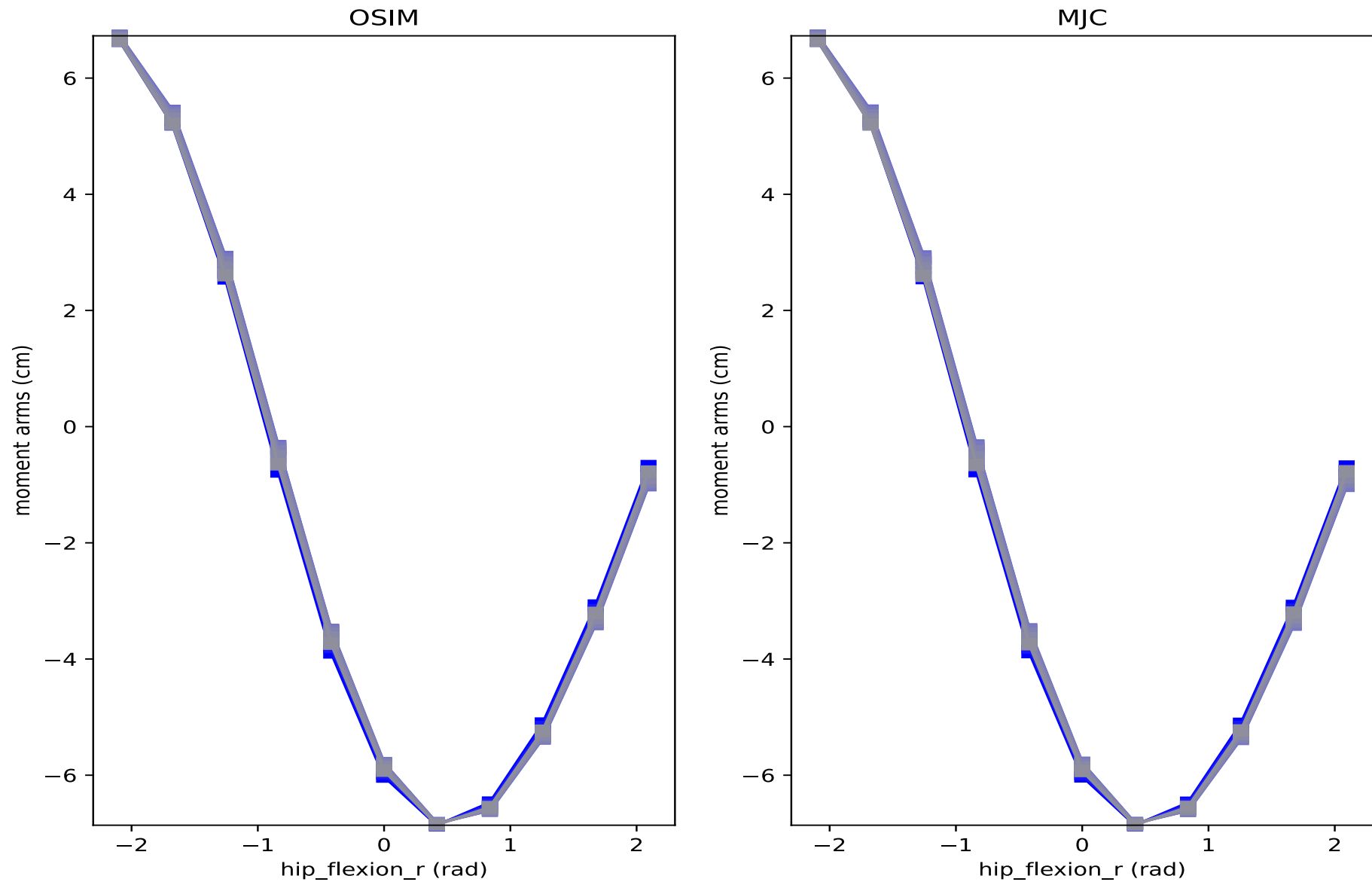
Overall comparison of muscle moment arms before/after optimization

Moment arm comparison of all muscles (cm)



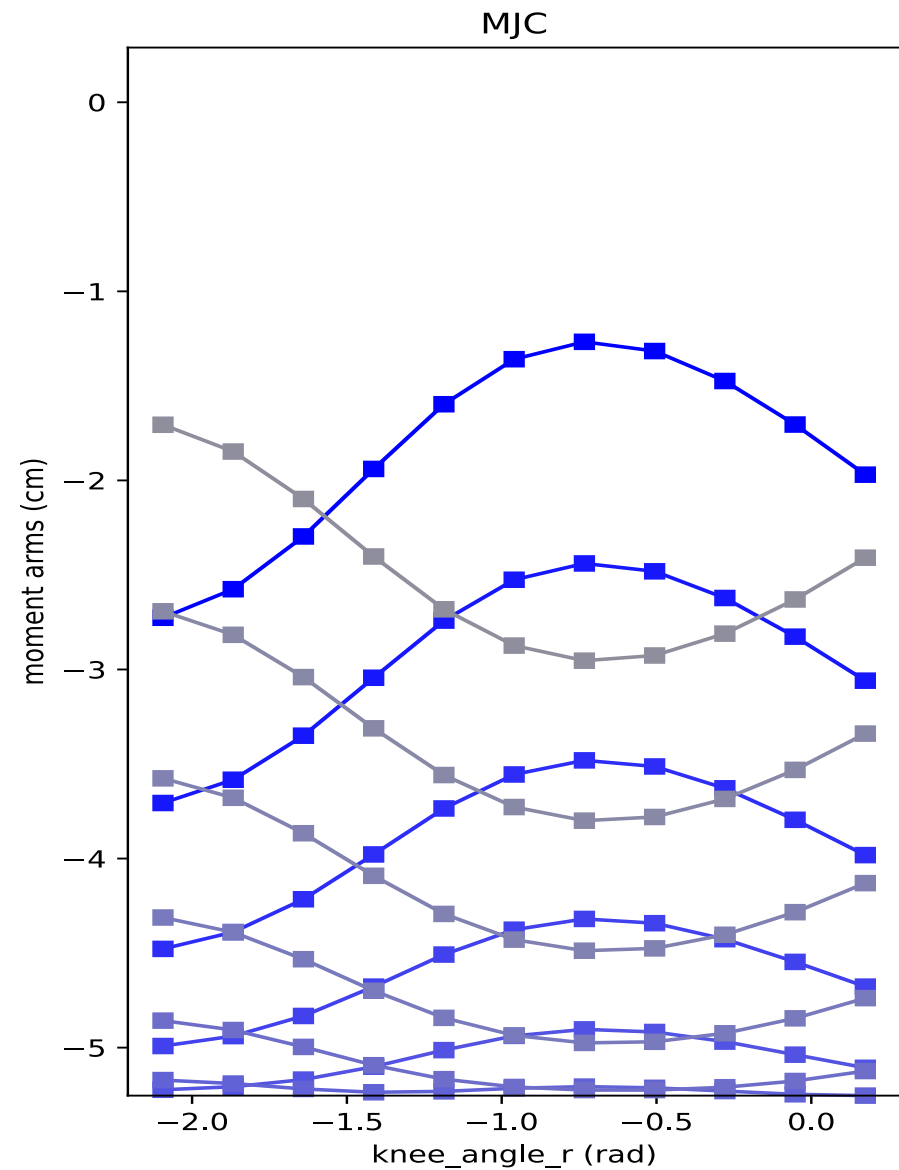
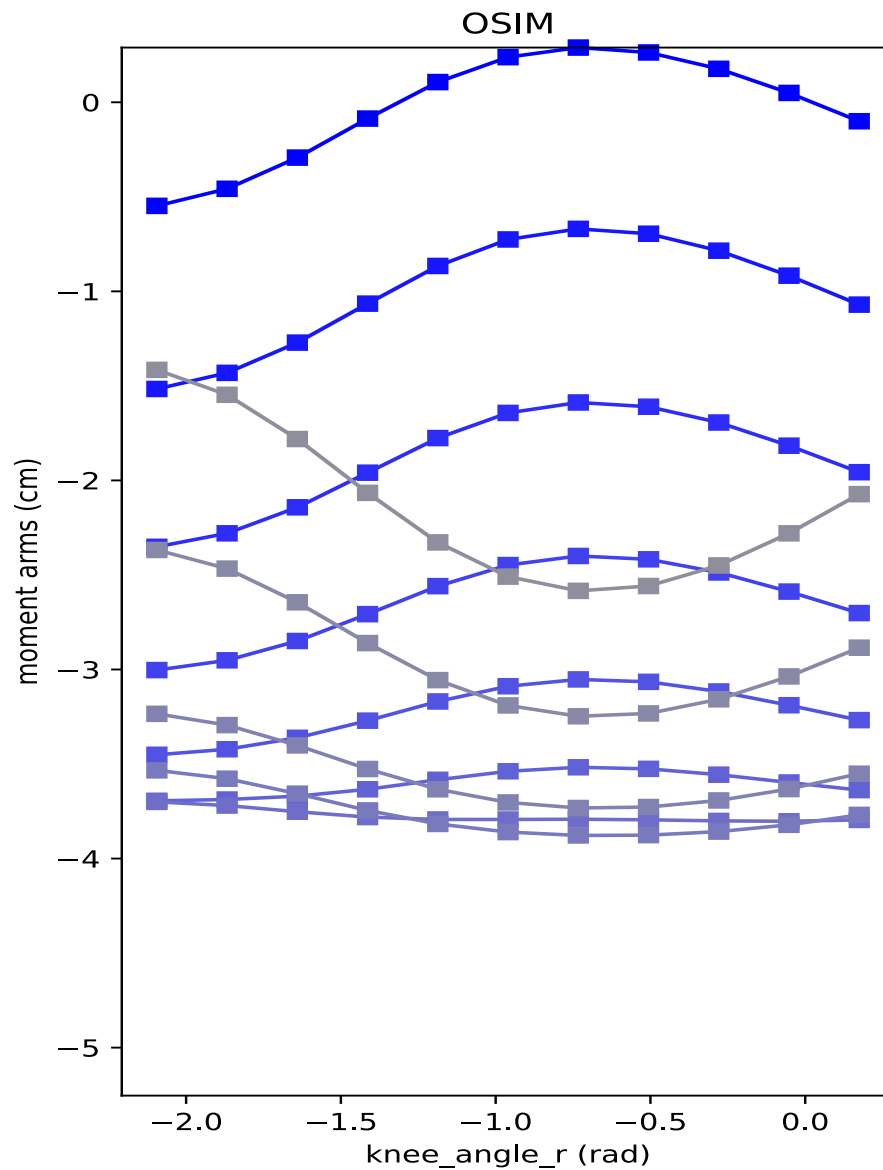
Muscle specific moment arm comparison with Osim model after optimization

bifemlh_r - hip_flexion_r - knee_angle_r



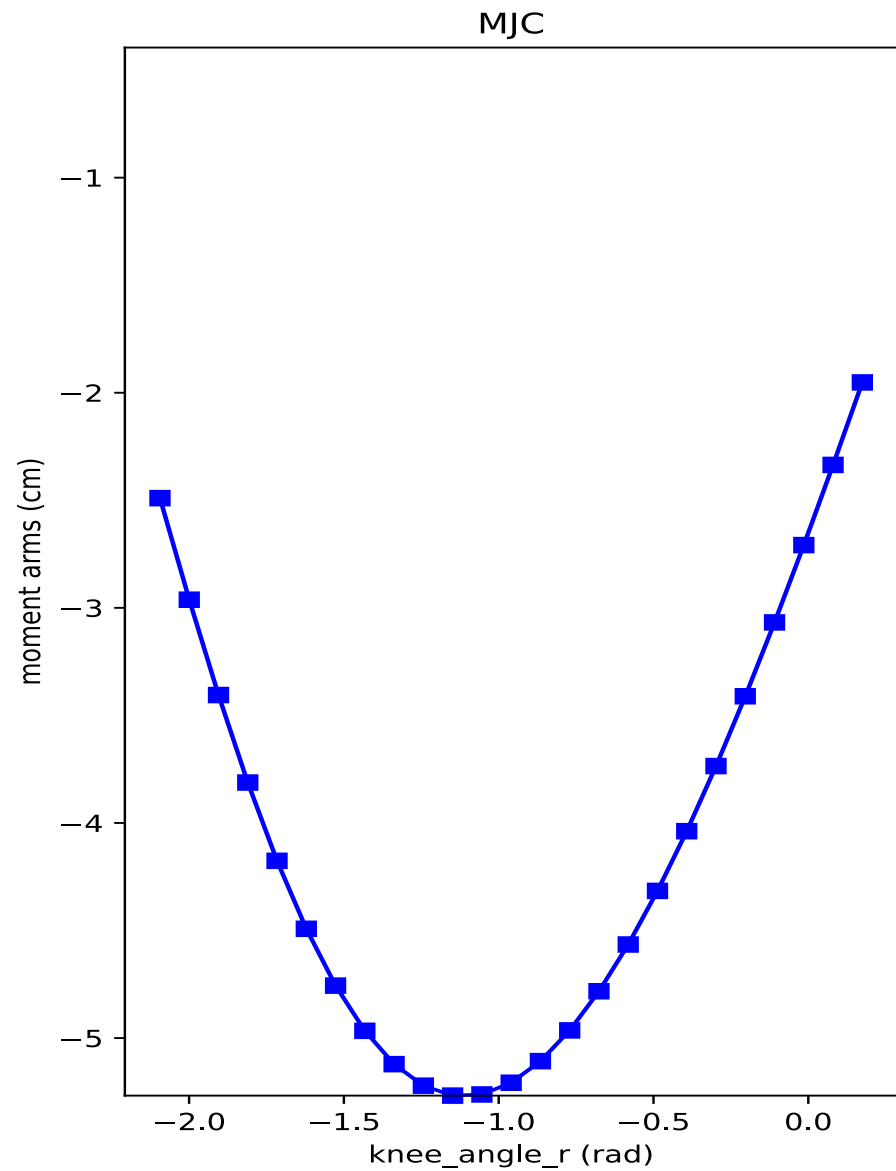
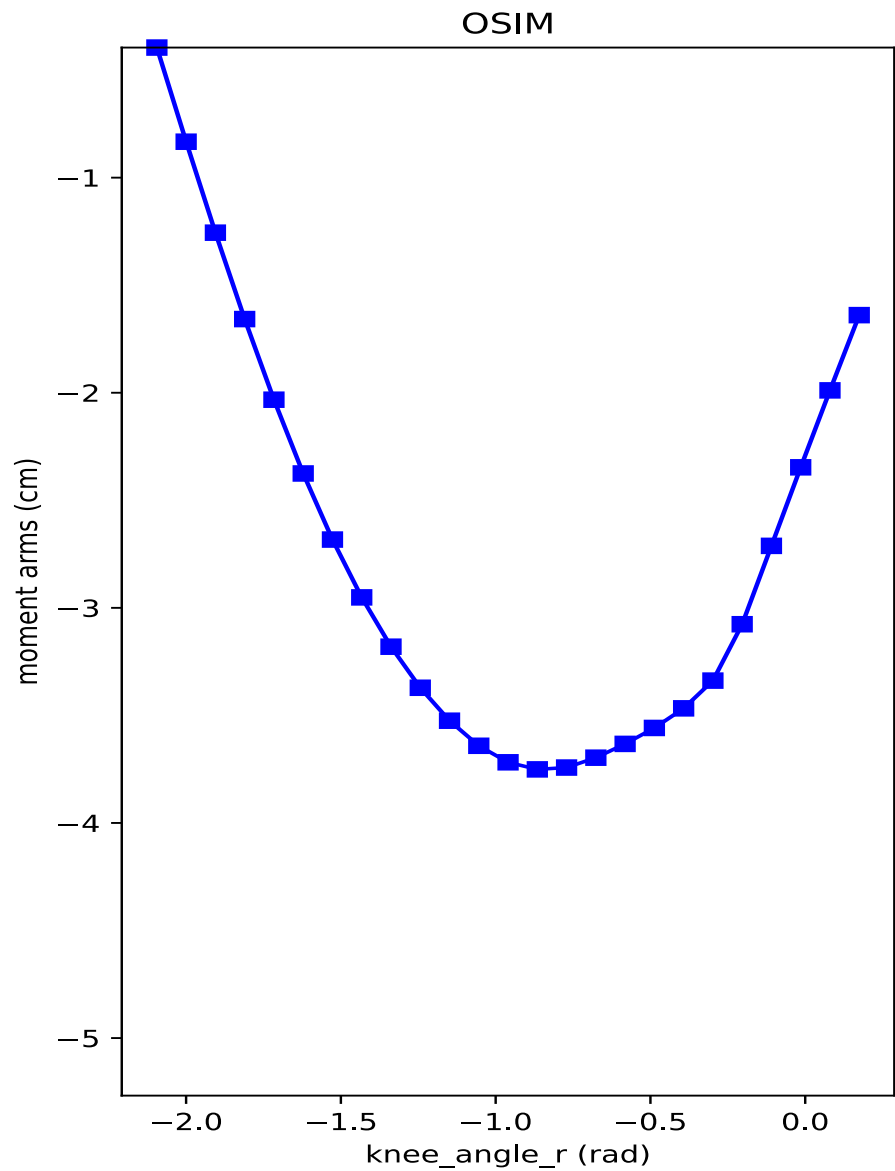
Muscle specific moment arm comparison with Osim model after optimization

bifemlh_r - hip_flexion_r - knee_angle_r



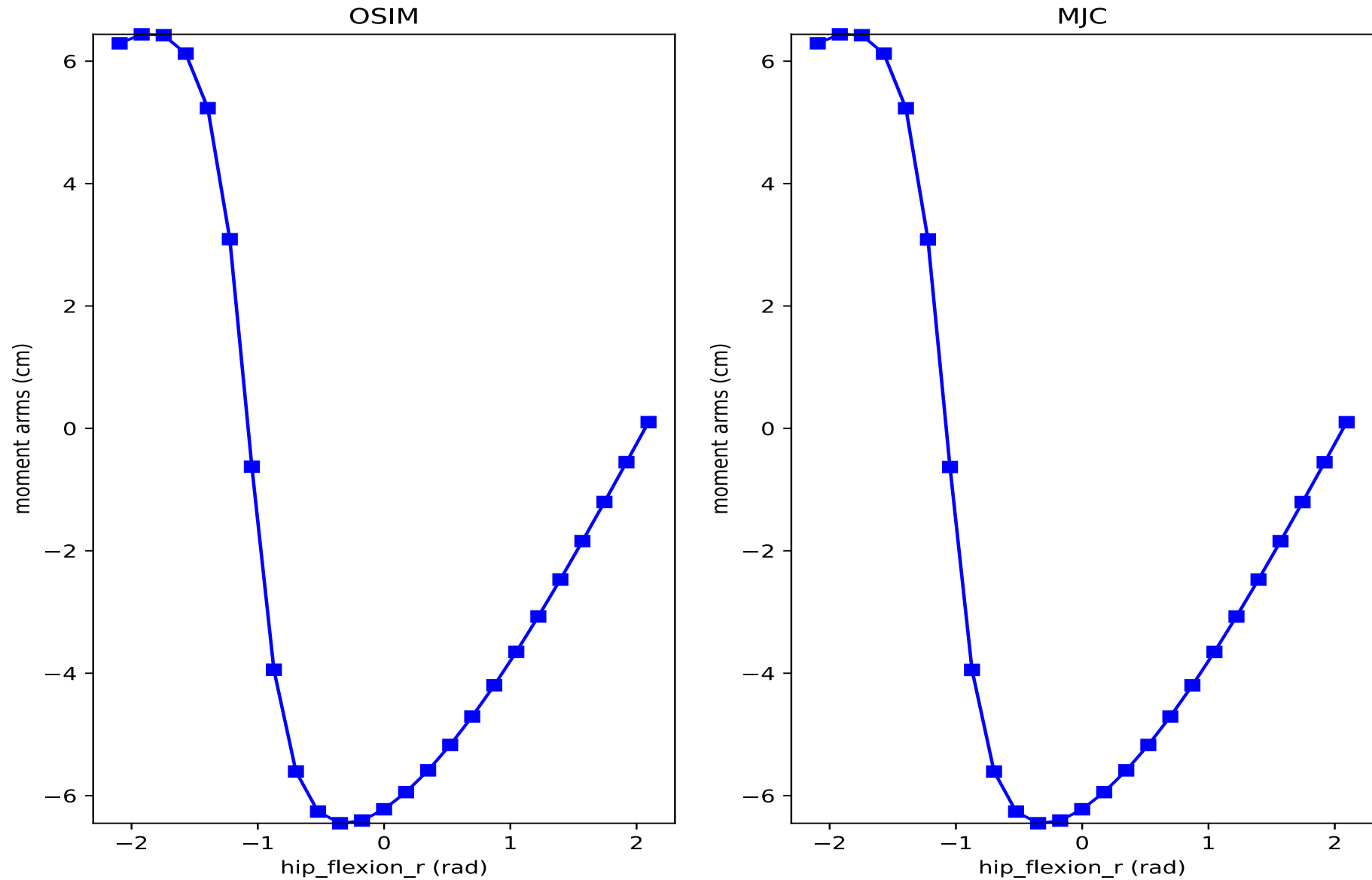
Muscle specific moment arm comparison with Osim model after optimization

bifemsh_r - knee_angle_r



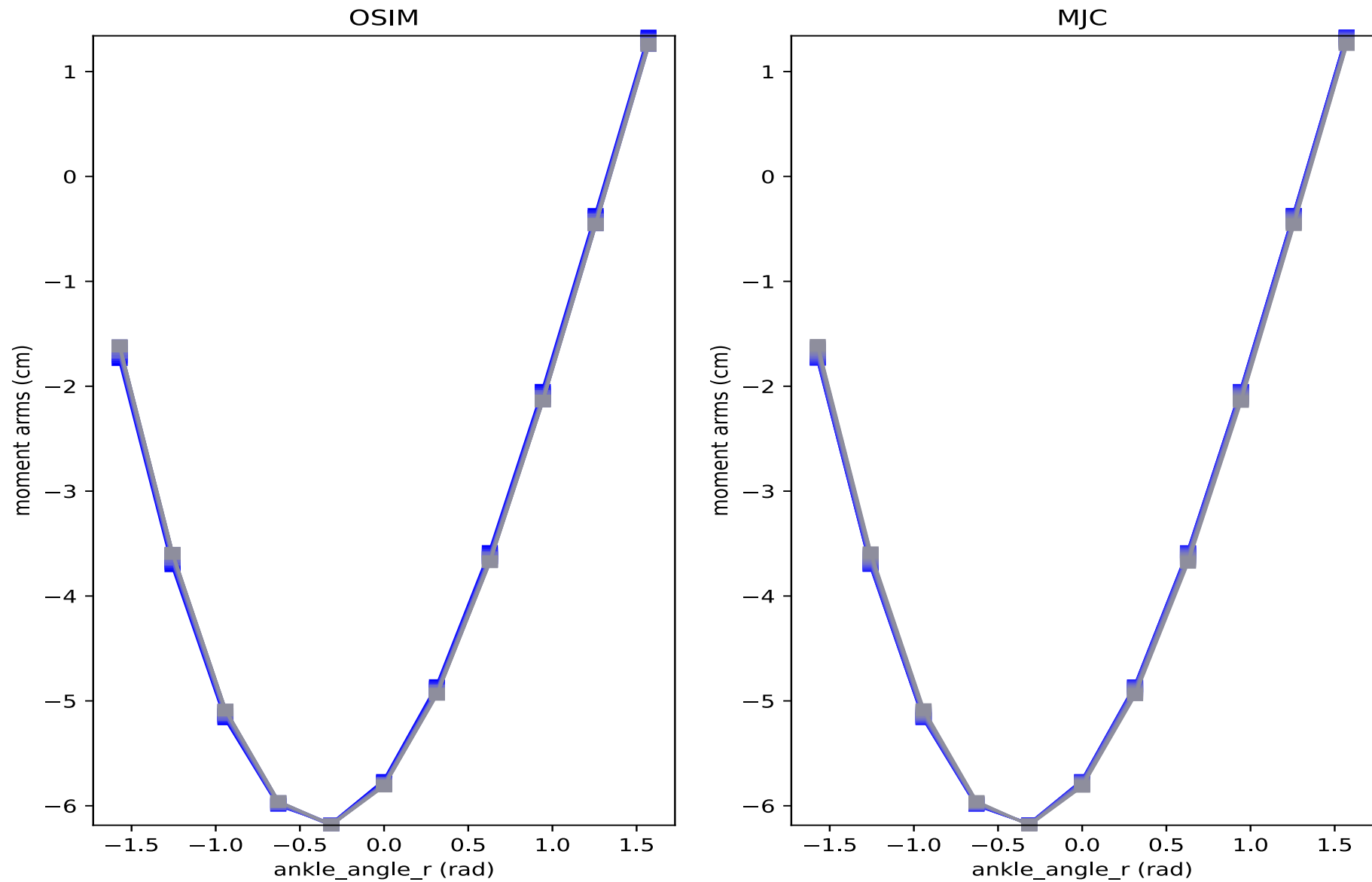
Muscle specific moment arm comparison with Osim model after optimization

glut_max2_r - hip_flexion_r



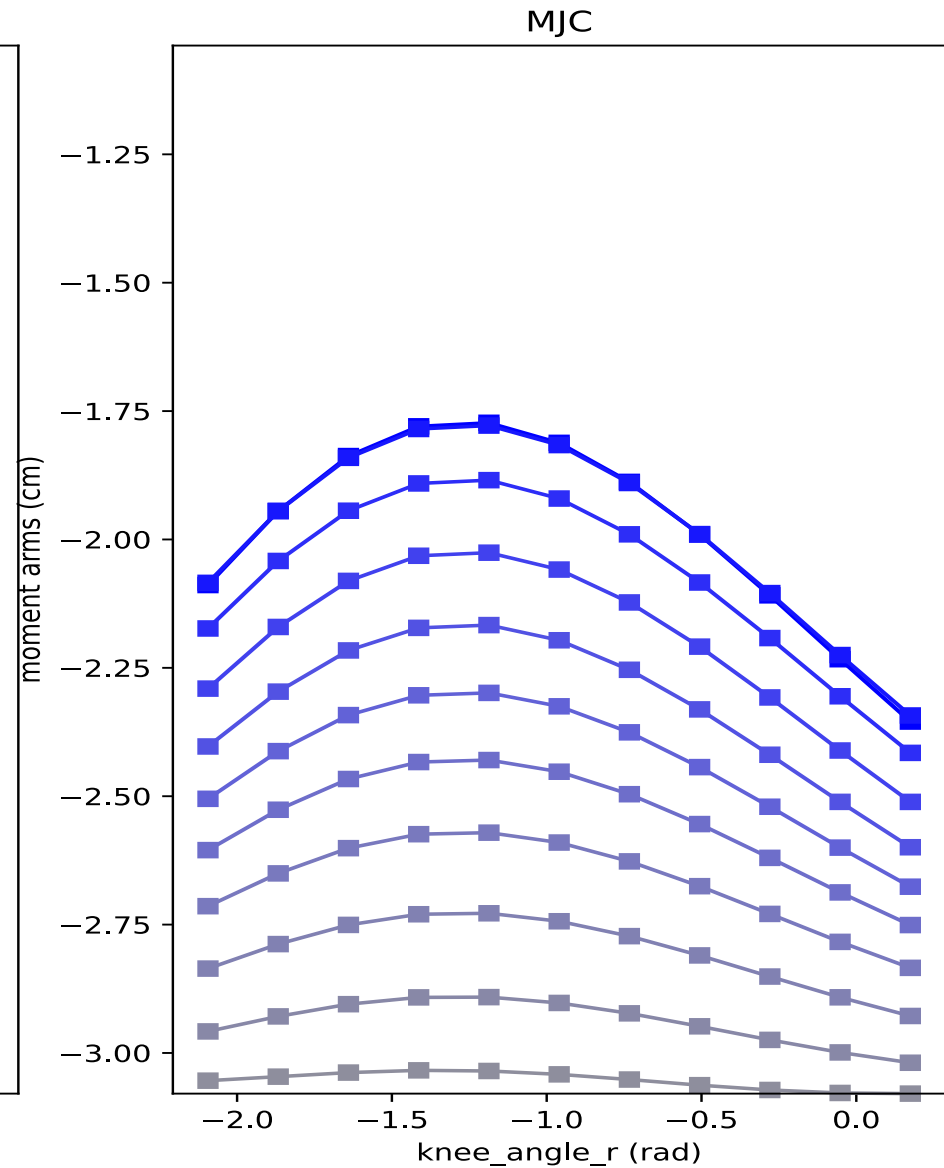
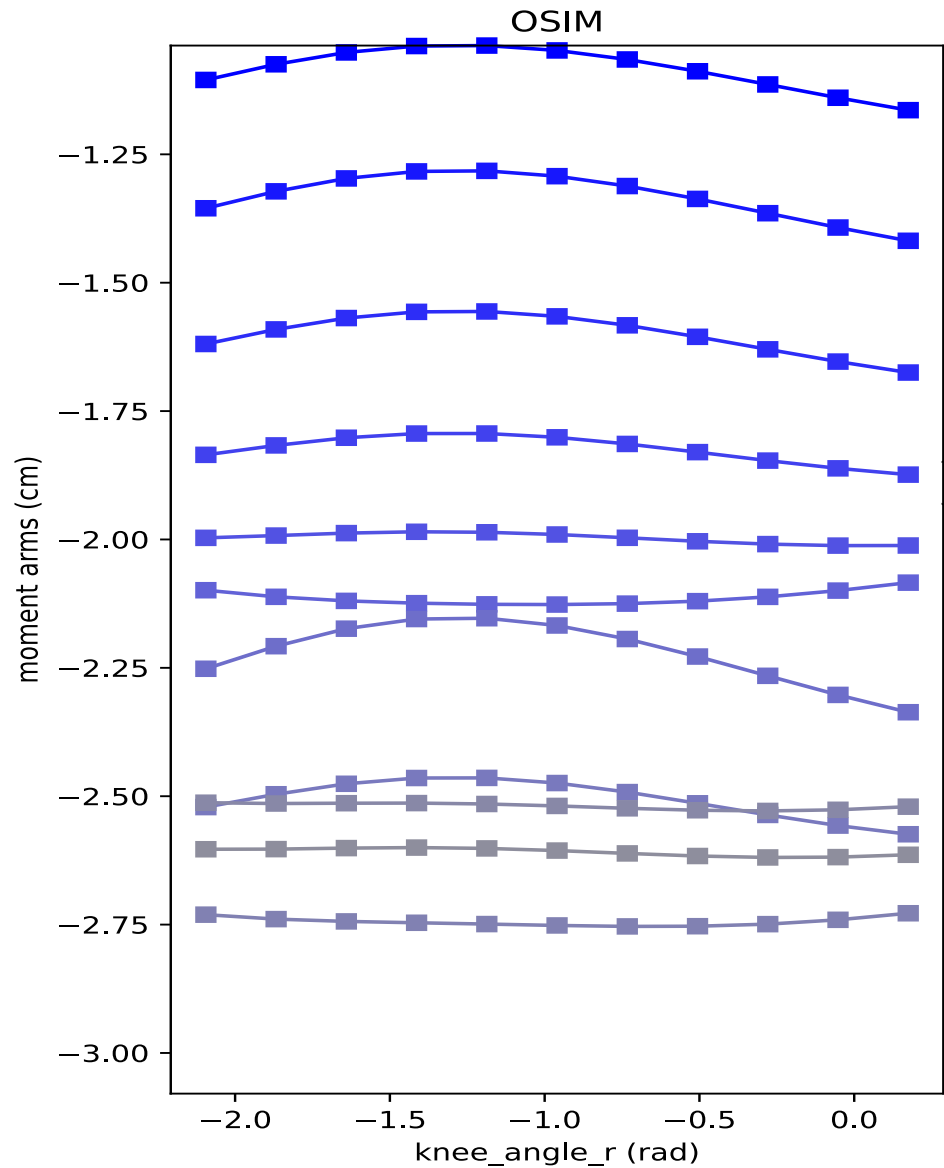
Muscle specific moment arm comparison with Osim model after optimization

med_gas_r - ankle_angle_r - knee_angle_r



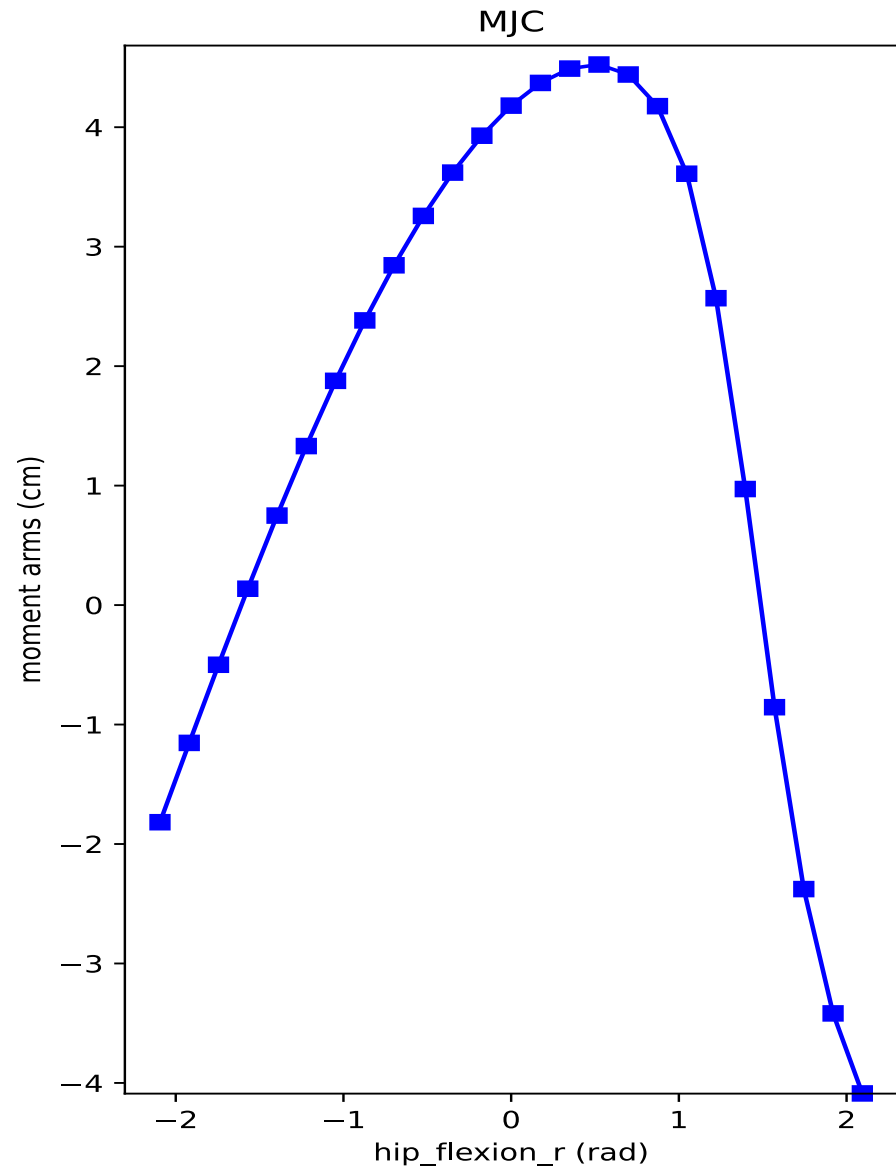
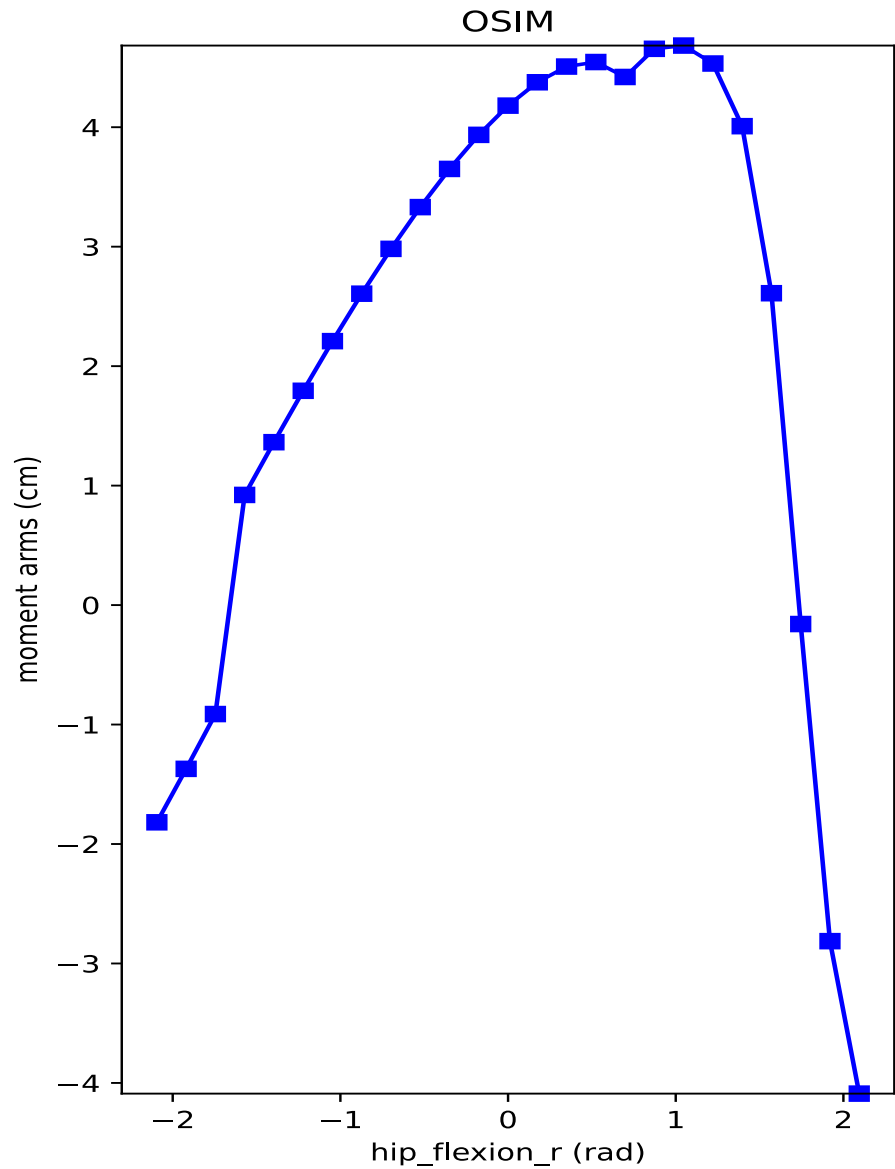
Muscle specific moment arm comparison with Osim model after optimization

med_gas_r - ankle_angle_r - knee_angle_r



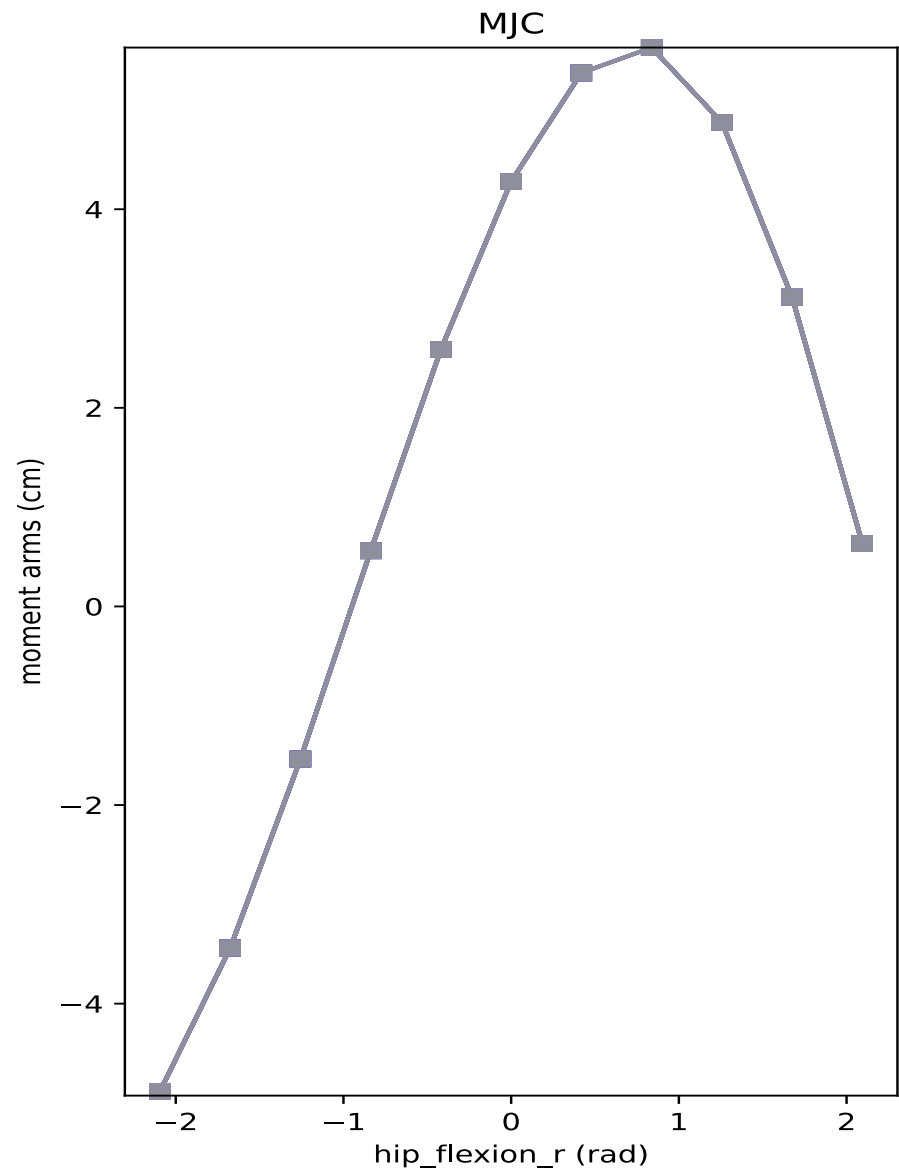
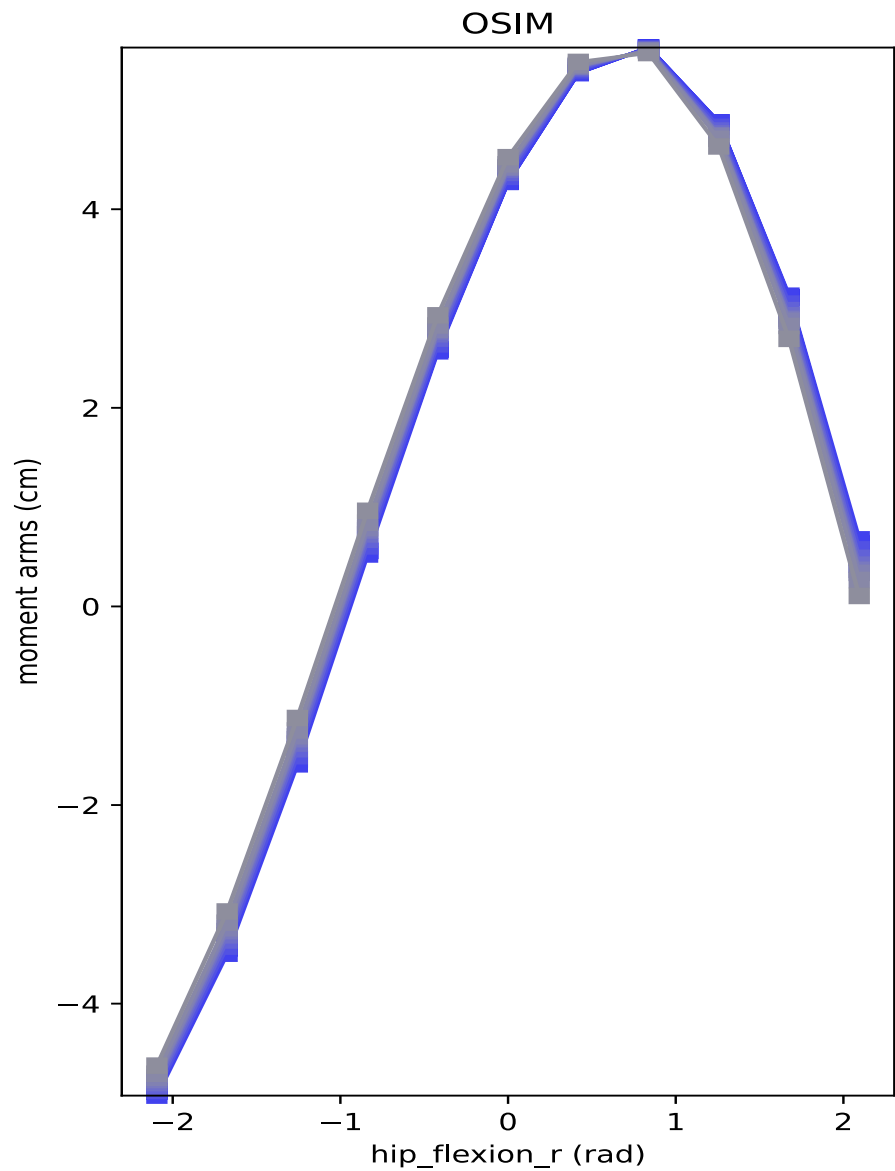
Muscle specific moment arm comparison with Osim model after optimization

psoas_r - hip_flexion_r



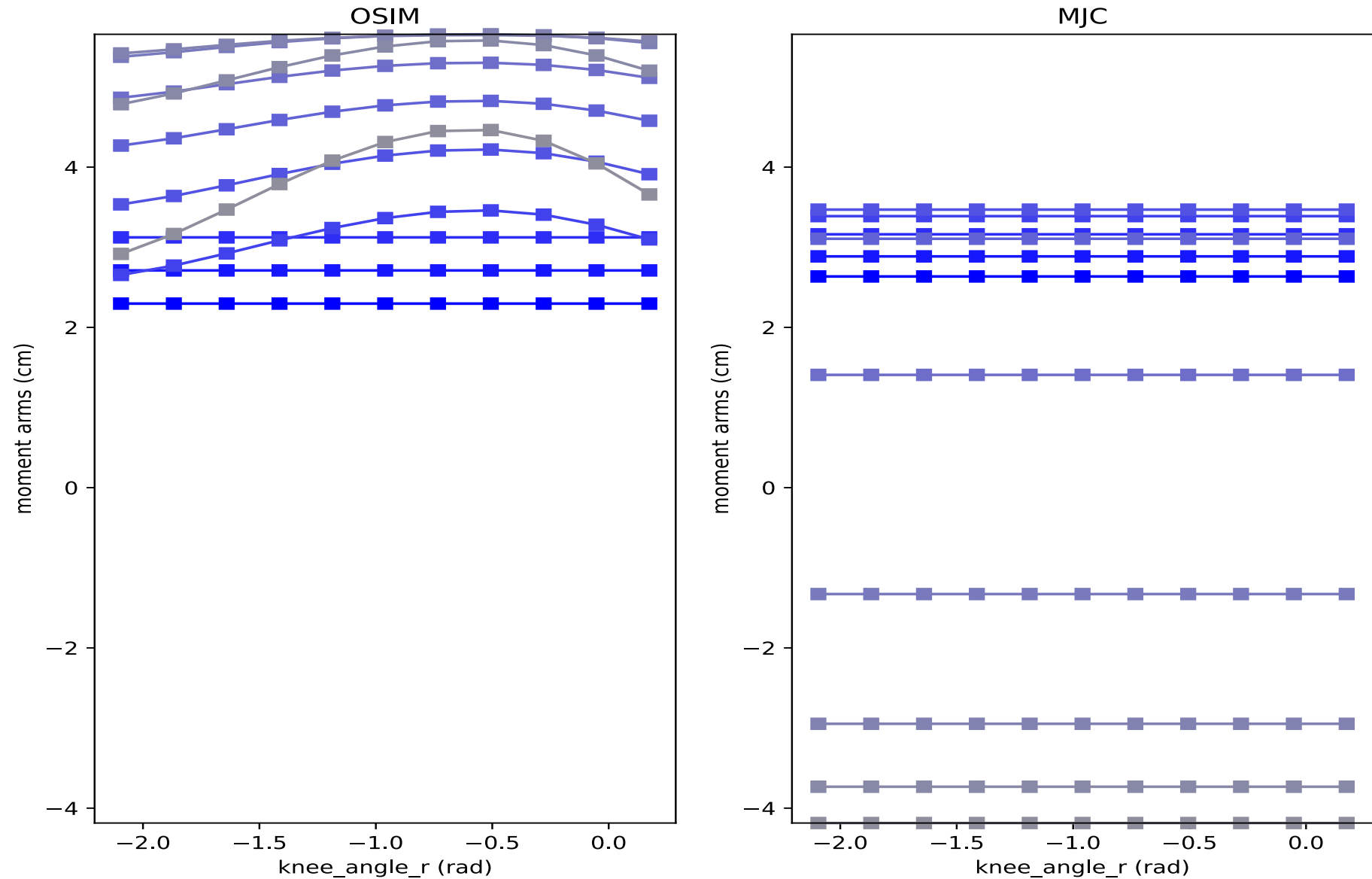
Muscle specific moment arm comparison with Osim model after optimization

rect_fem_r - hip_flexion_r - knee_angle_r



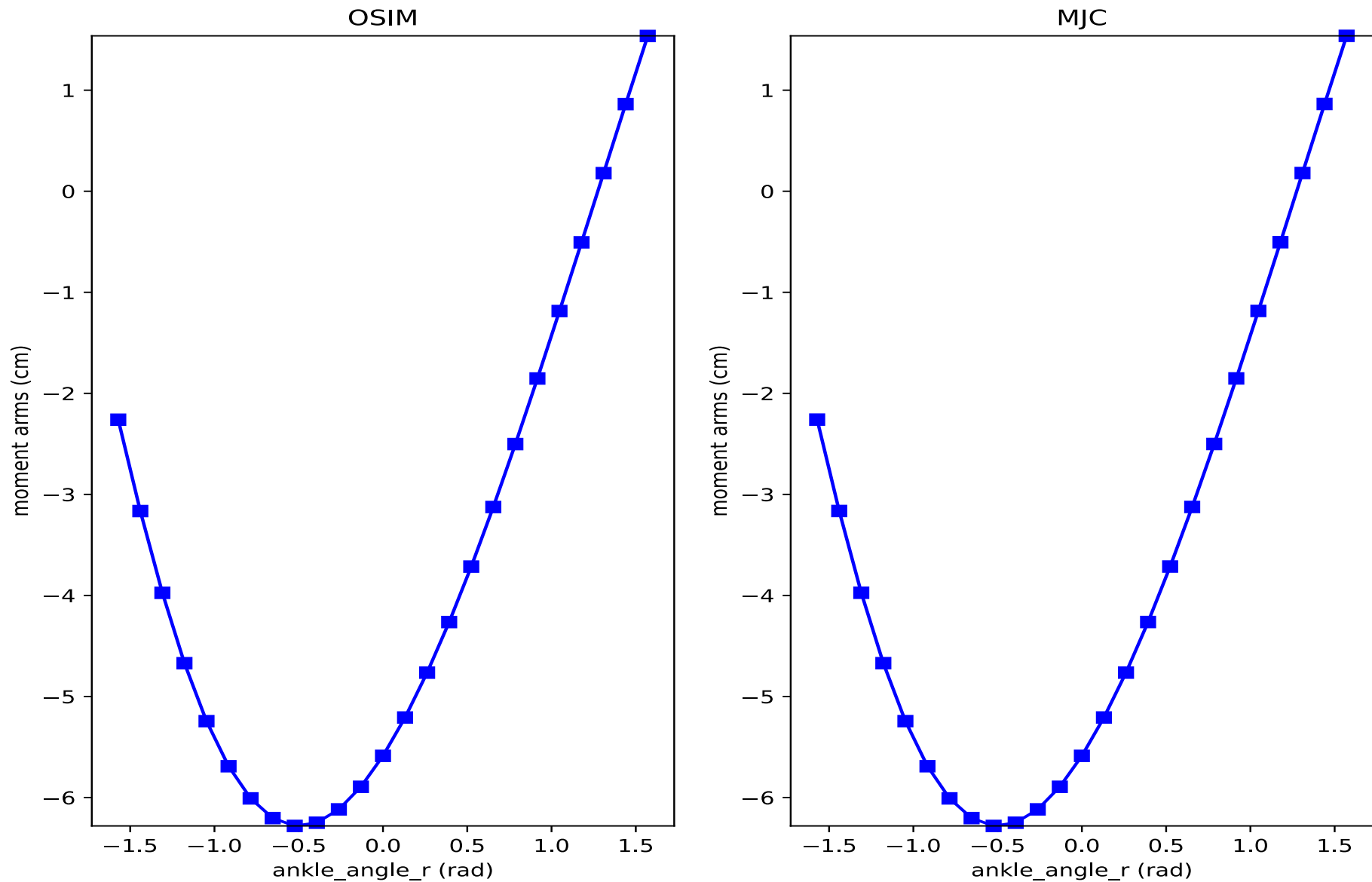
Muscle specific moment arm comparison with Osim model after optimization

rect_fem_r - hip_flexion_r - knee_angle_r



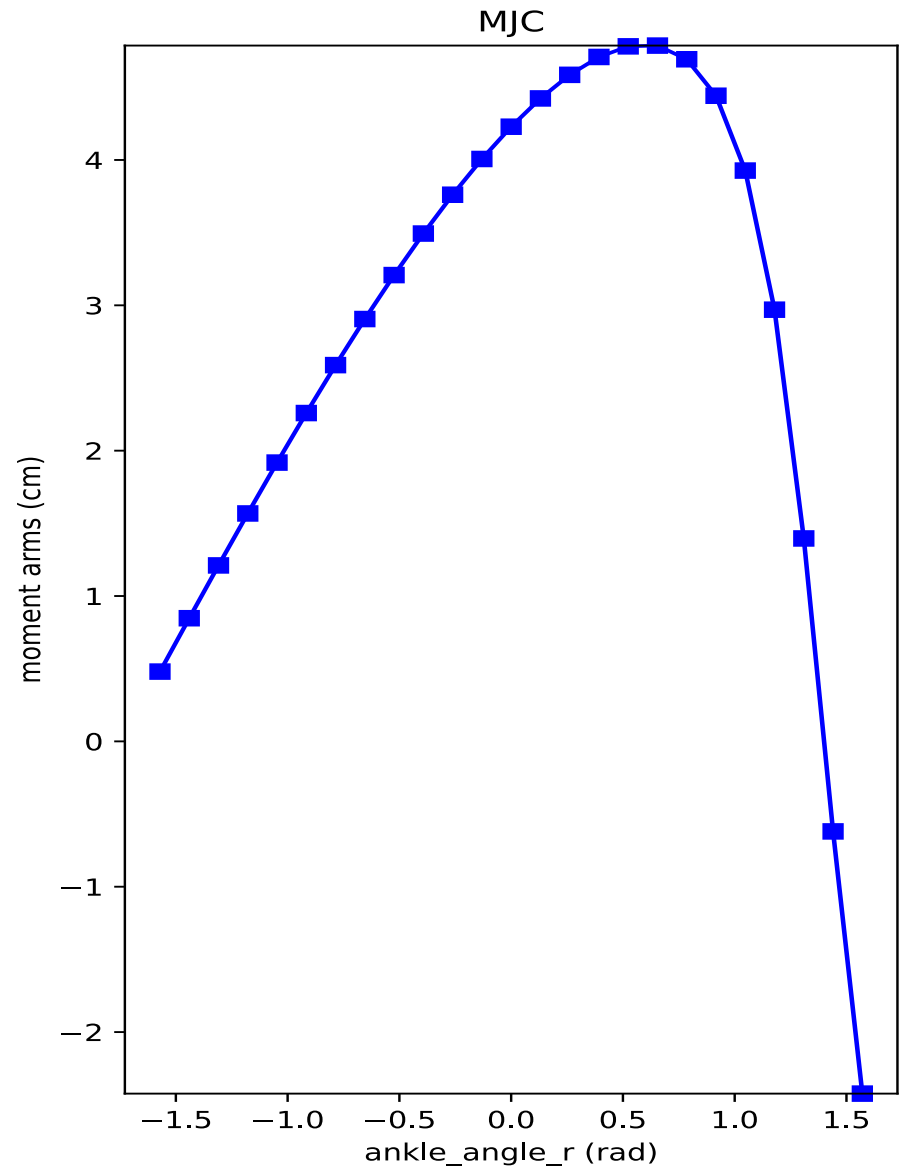
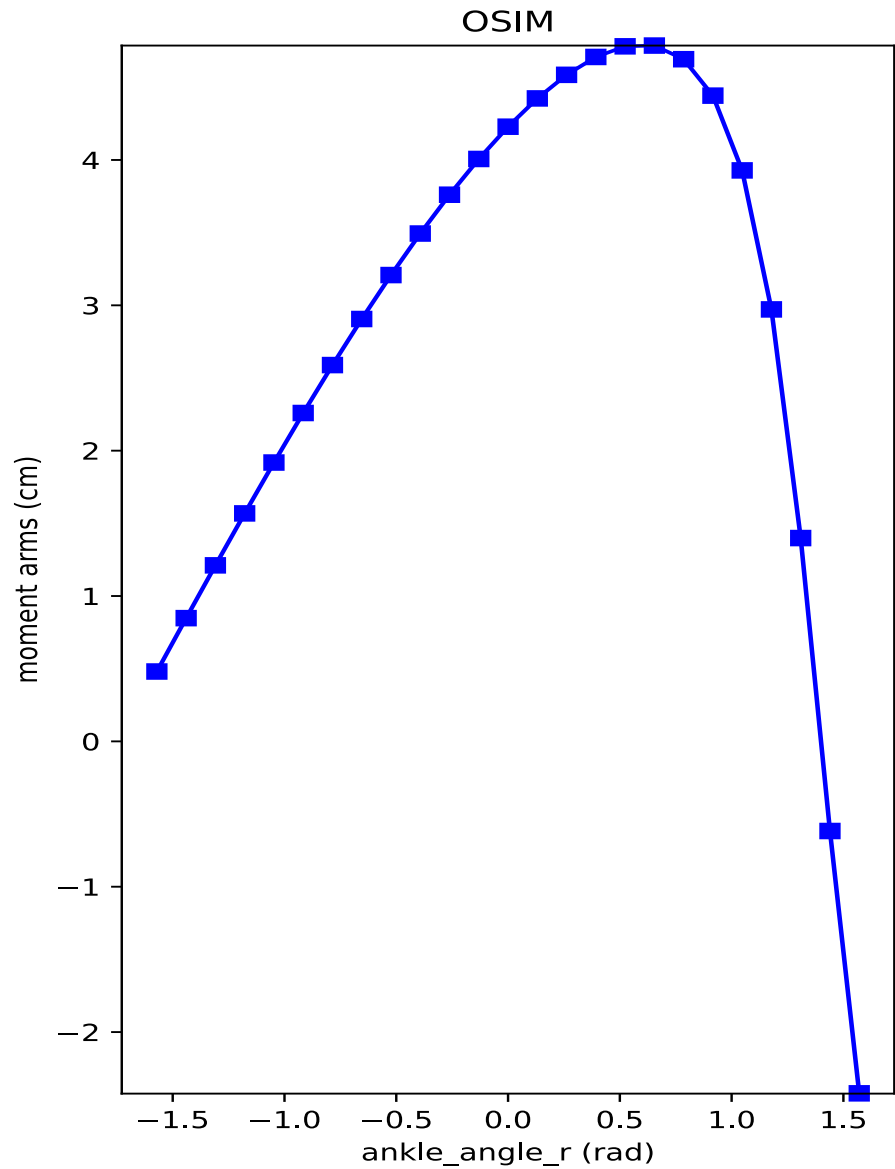
Muscle specific moment arm comparison with Osim model after optimization

soleus_r - ankle_angle_r



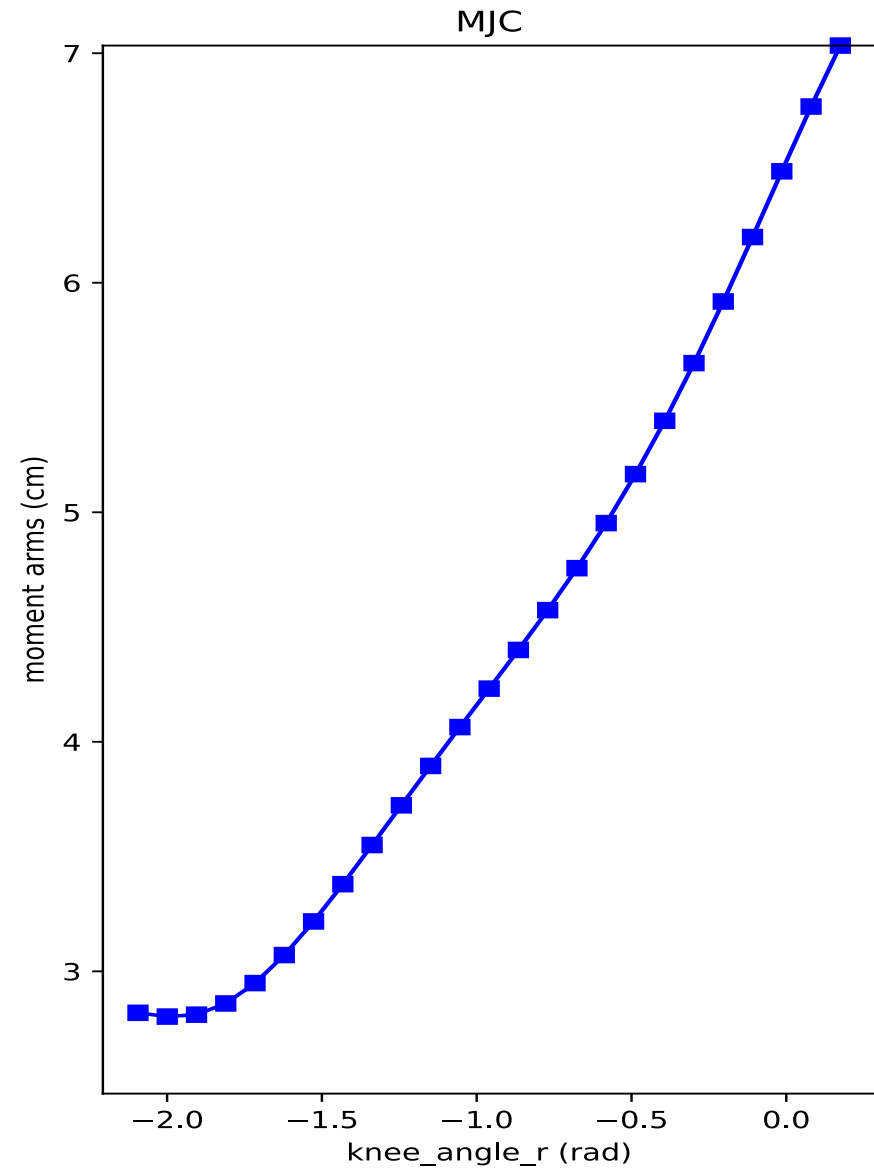
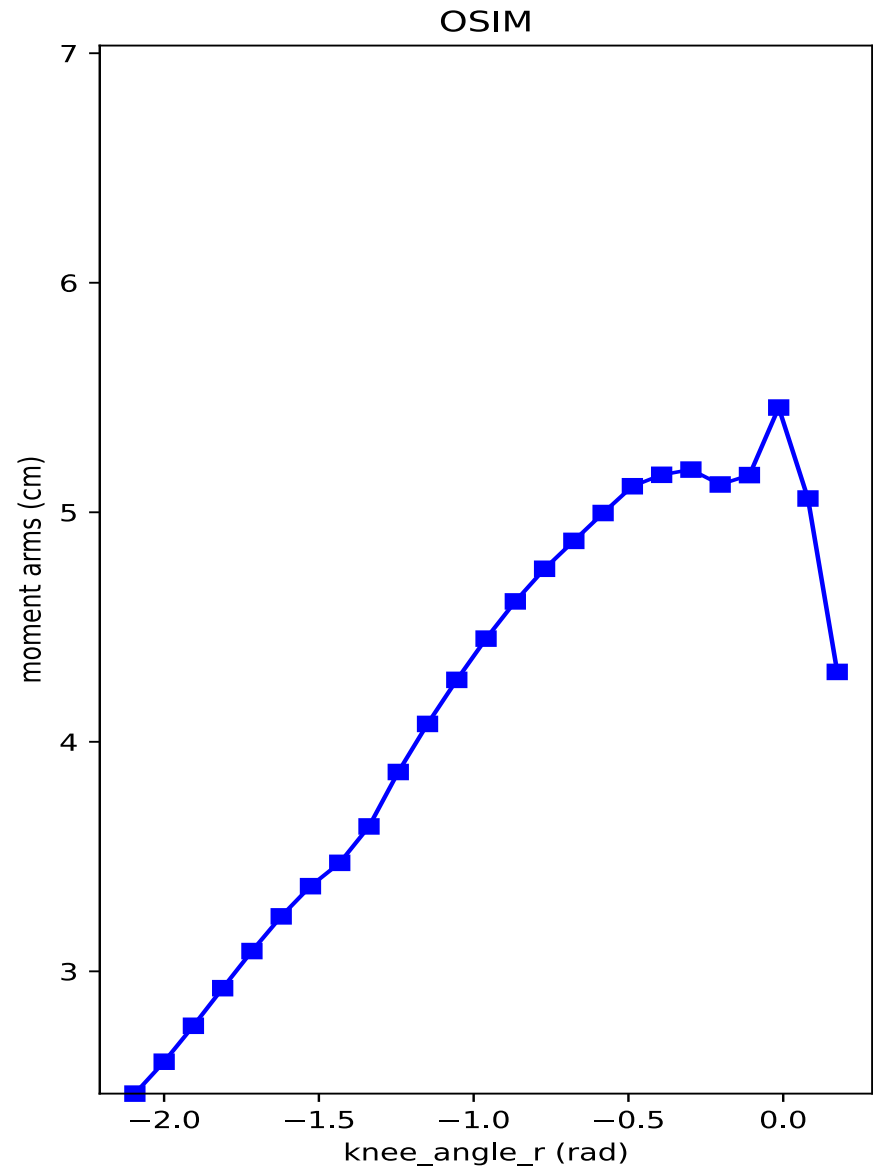
Muscle specific moment arm comparison with Osim model after optimization

tib_ant_r - ankle_angle_r



Muscle specific moment arm comparison with Osim model after optimization

vas_int_r - knee_angle_r



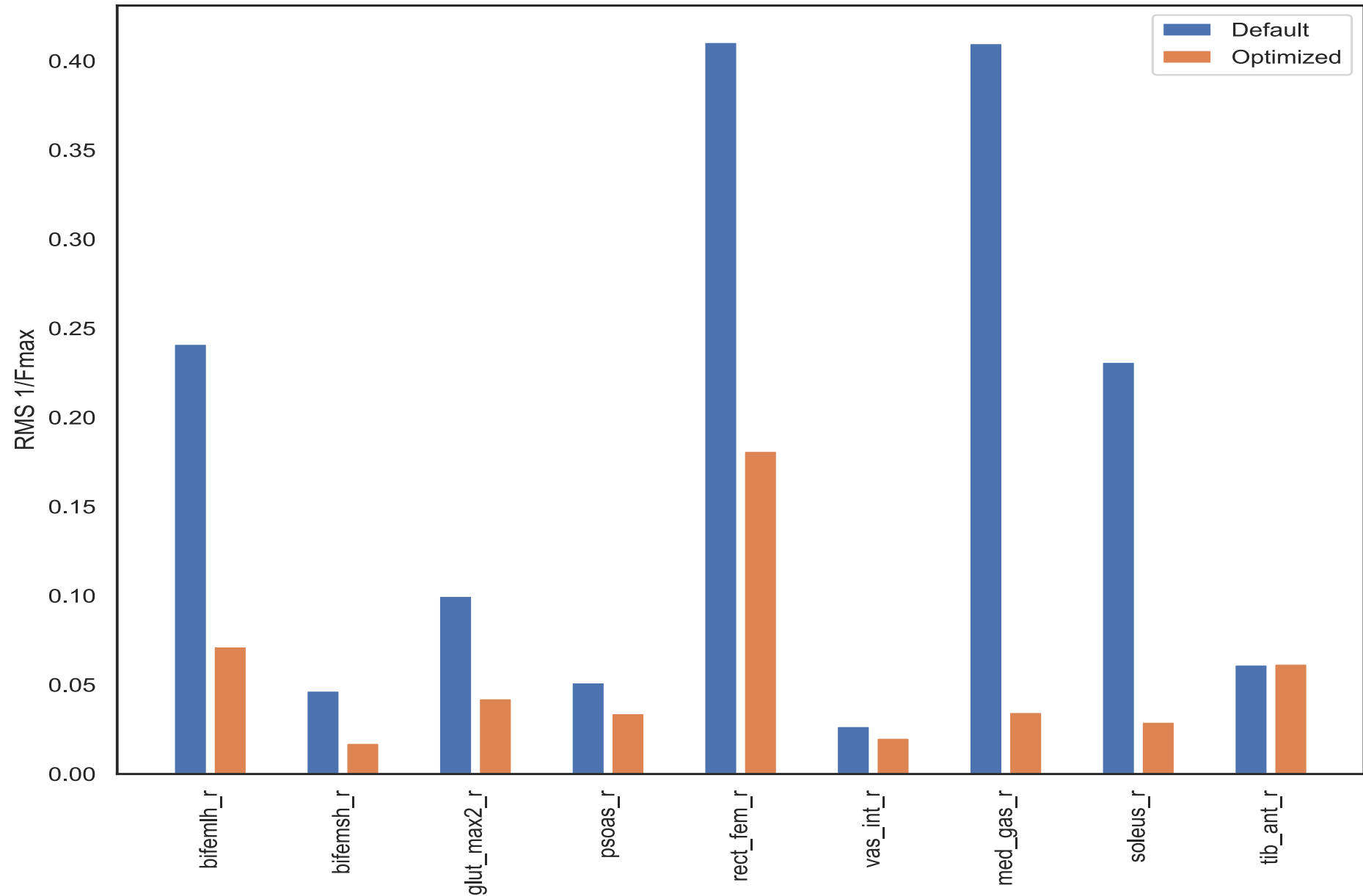
Step 3: Muscle Kinetic Validation

Muscle force-length property are compared between Osim and Mjc models. This force-length property only depends on muscle-fiber-tendon unit lengths. We made it isolated with the moment arm, so that the change in moment arms will not affect the muscle force properties. The muscle-fiber-tendon unit lengths were roughly even extracted (from shortest to longest) with all possible body postures. A bar plot of the force errors of all muscle before and after optimization is included. Then the detail force-length curve comparison plot of each muscle is included.

How to interpret the plot:

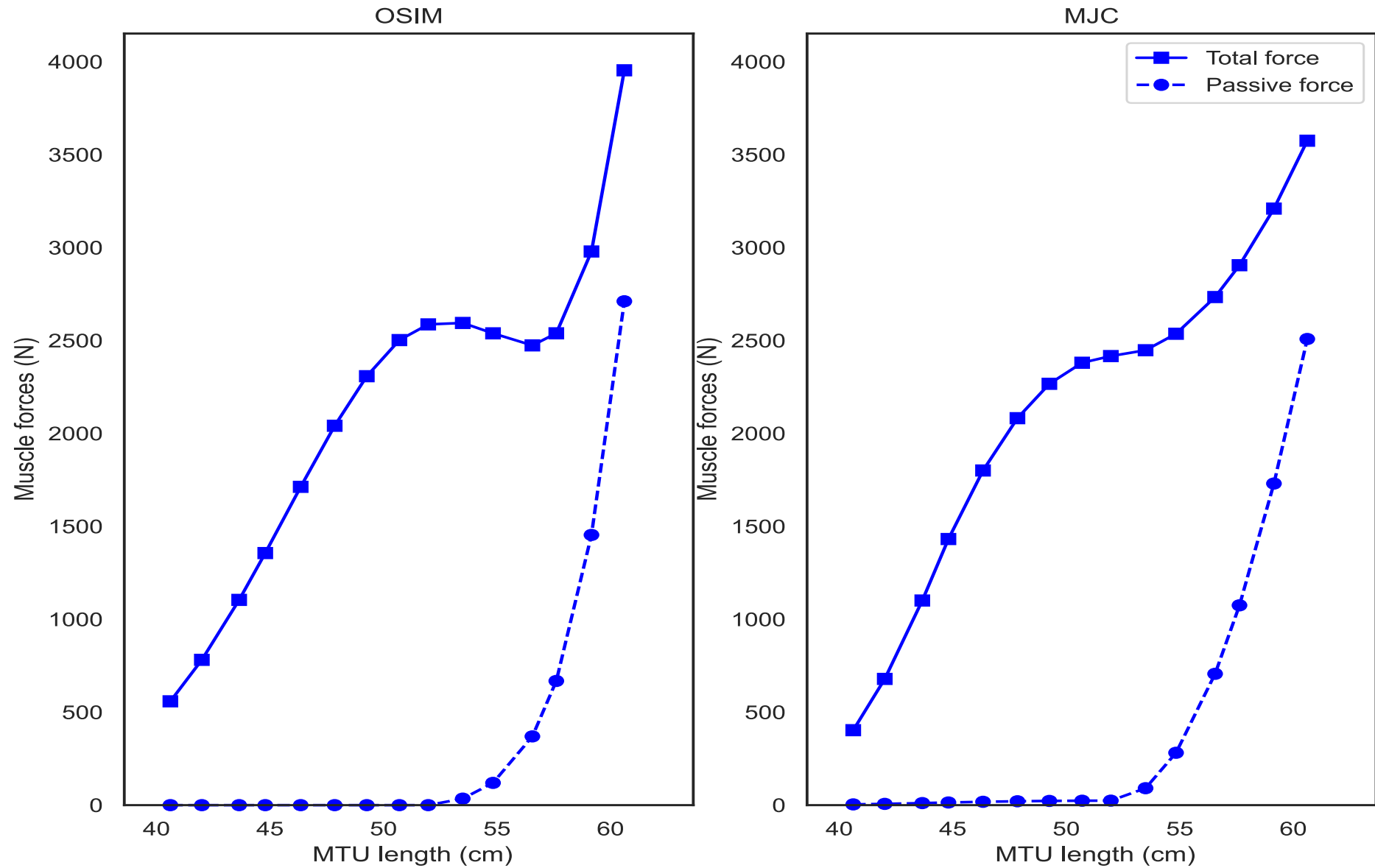
Global title indicate the muscle name. X axis indicate the muscle-fiber-tendon unit length. Y axis is the muscle force (unnormalized)

Overall comparison of muscle force-length relationship before/after optimization



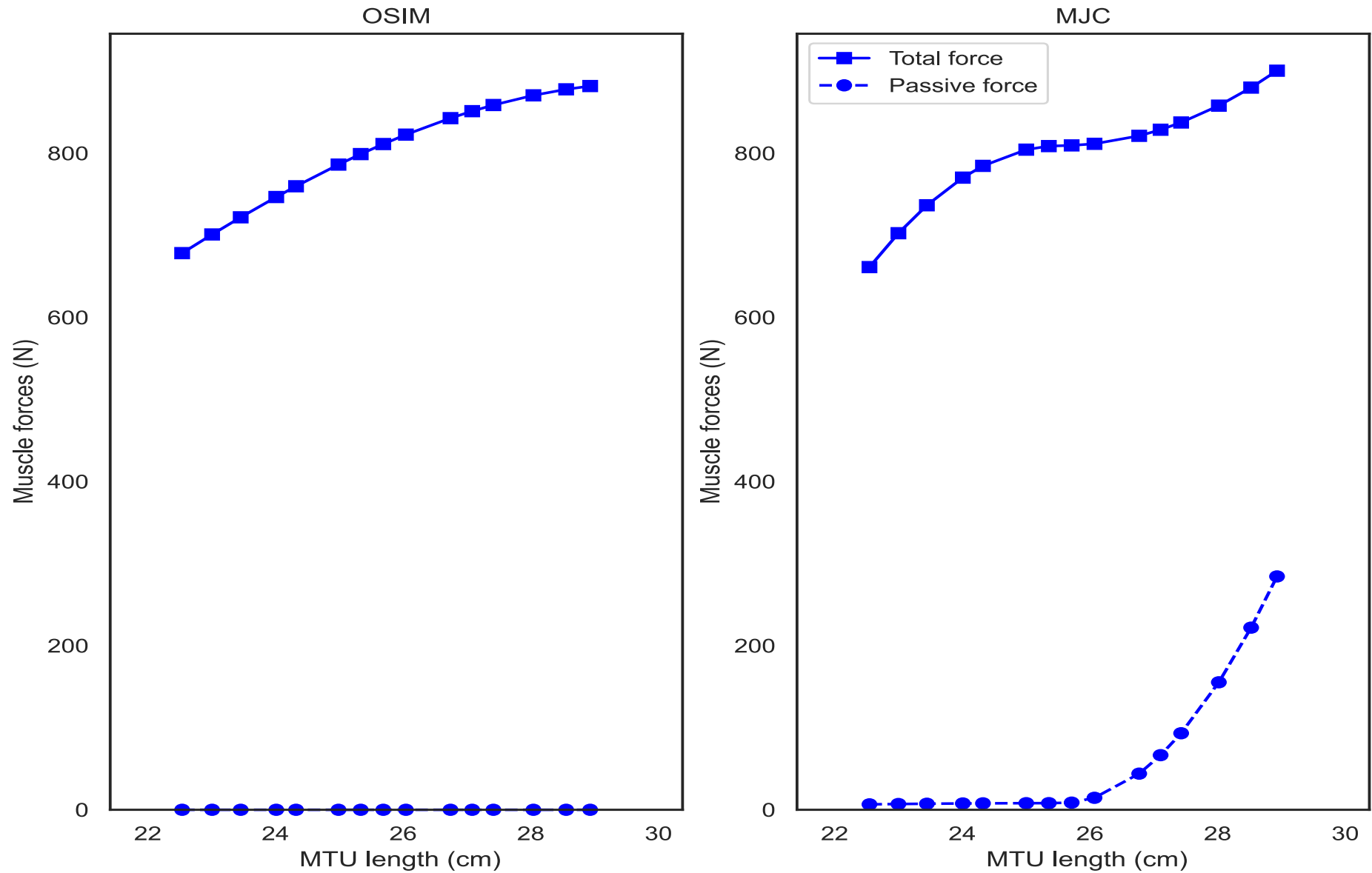
Muscle specific force comparison before/after optimization

bifemlh_r



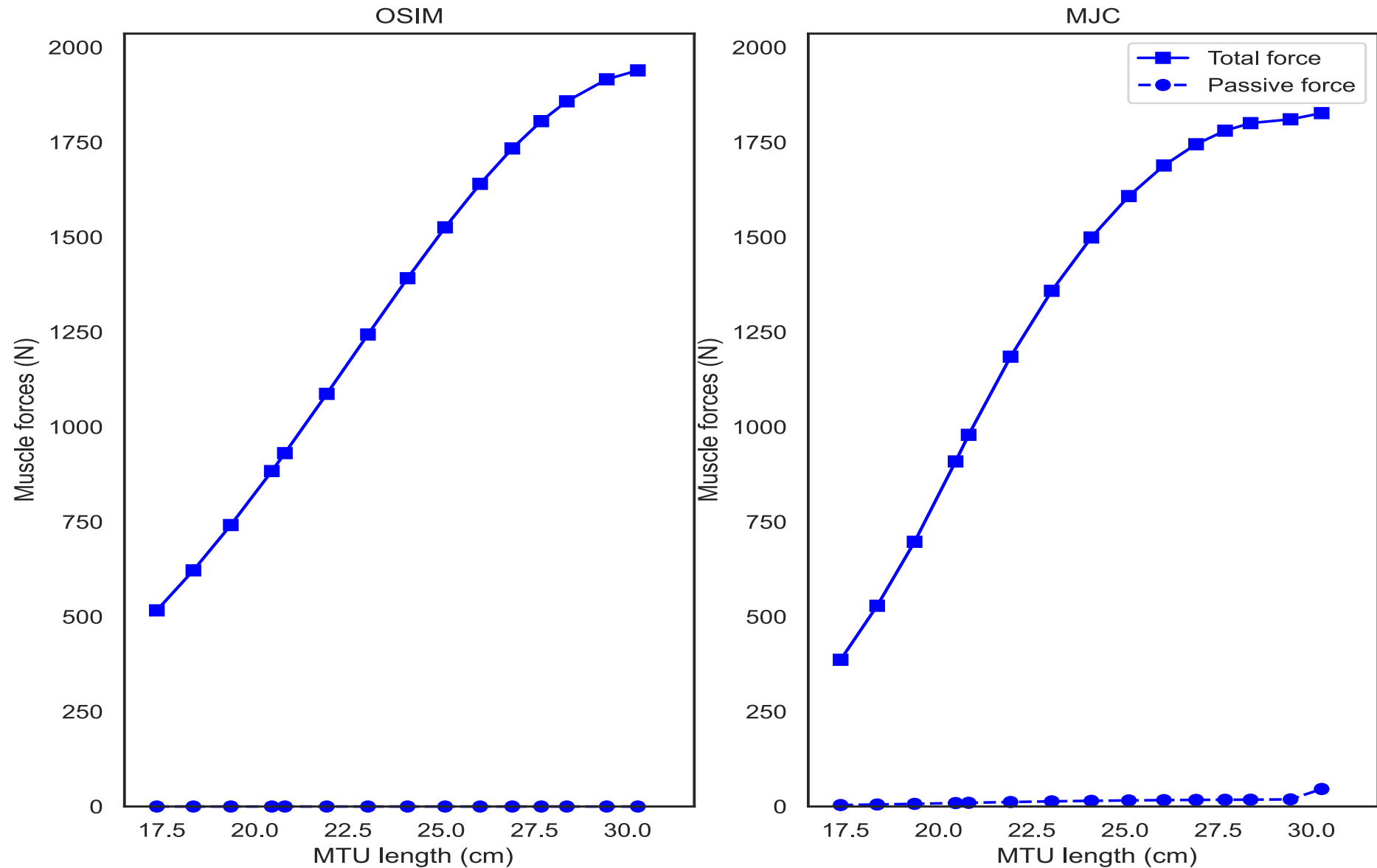
Muscle specific force comparison before/after optimization

bifemsh_r



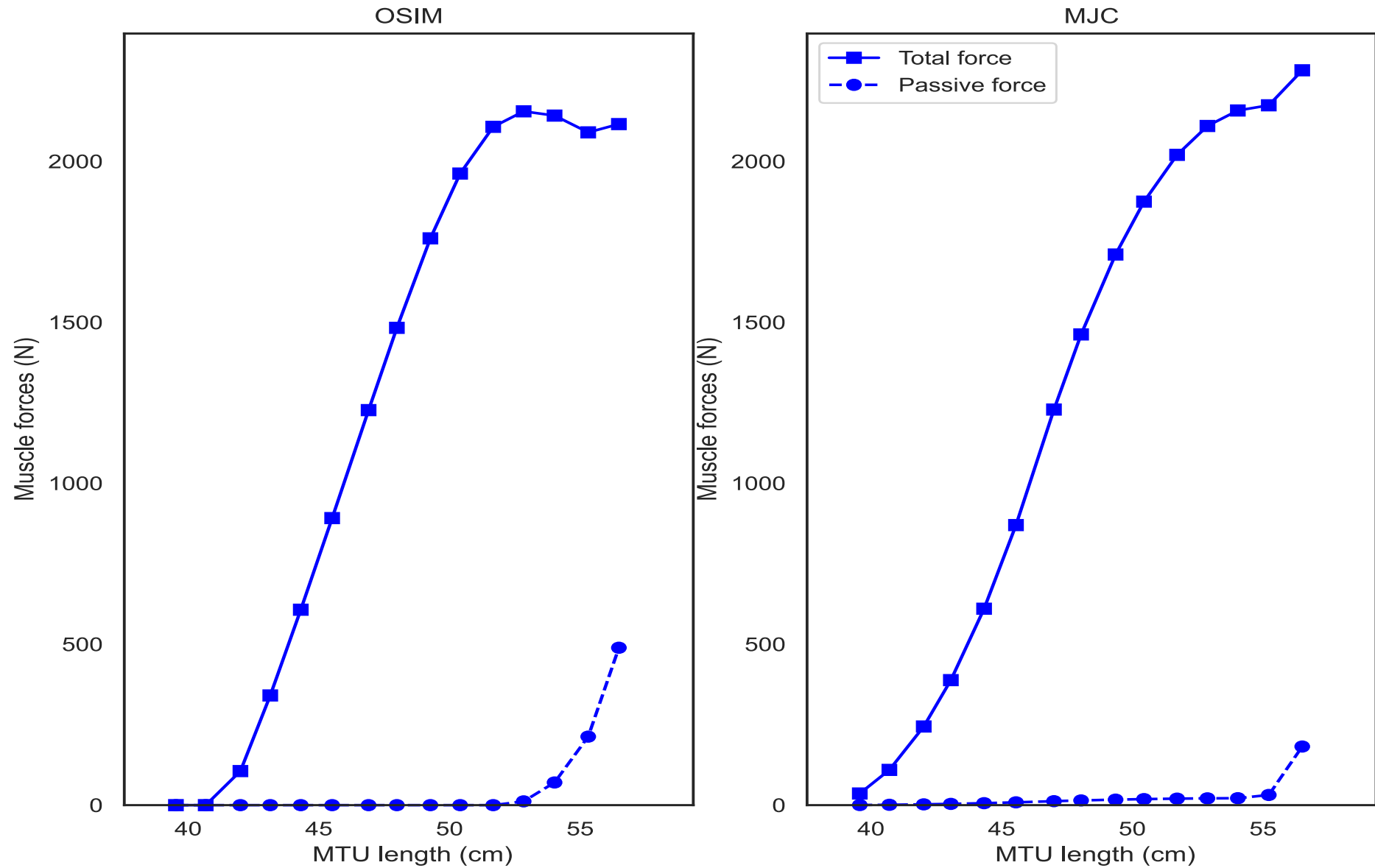
Muscle specific force comparison before/after optimization

glut_max2_r



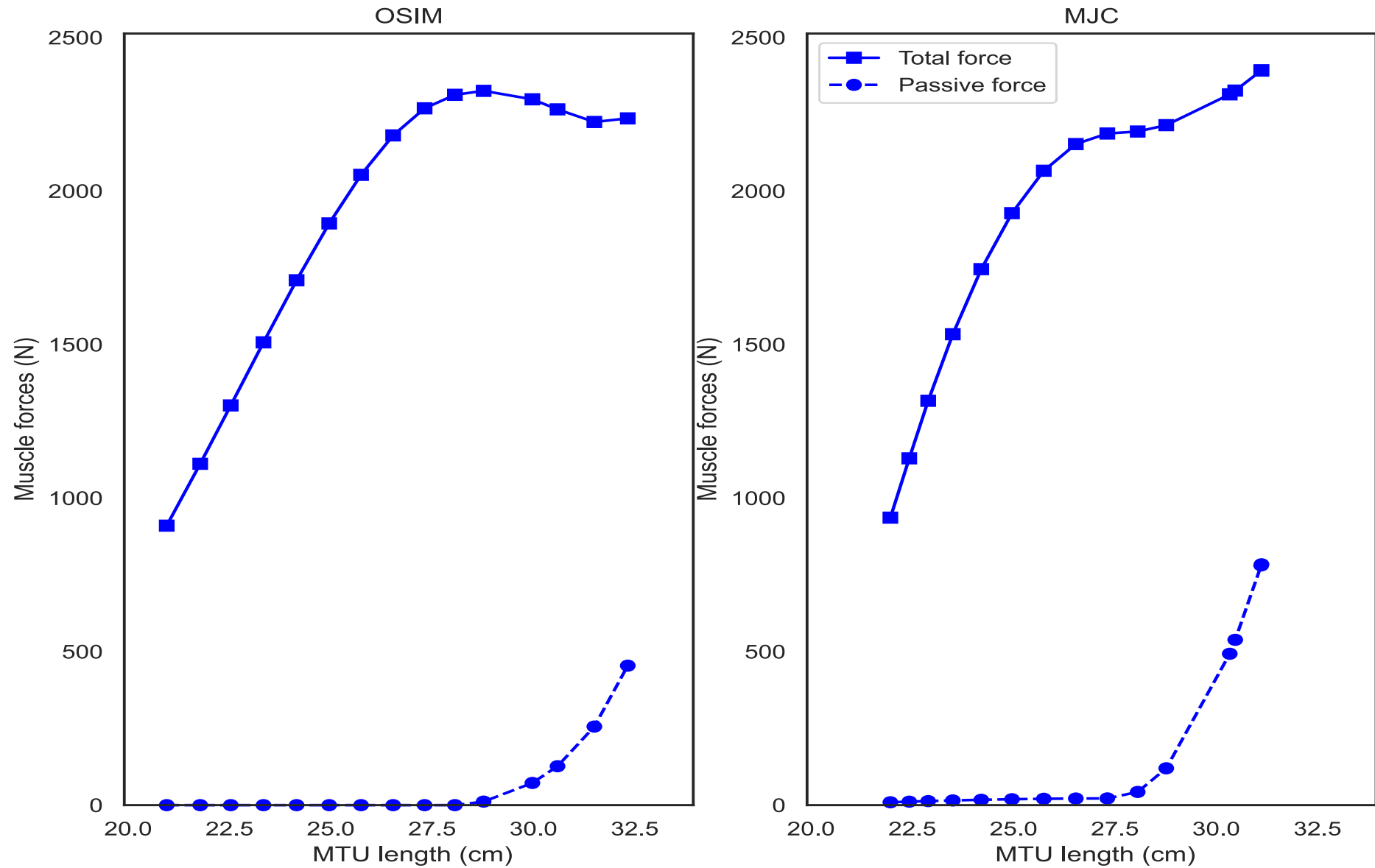
Muscle specific force comparison before/after optimization

med_gas_r



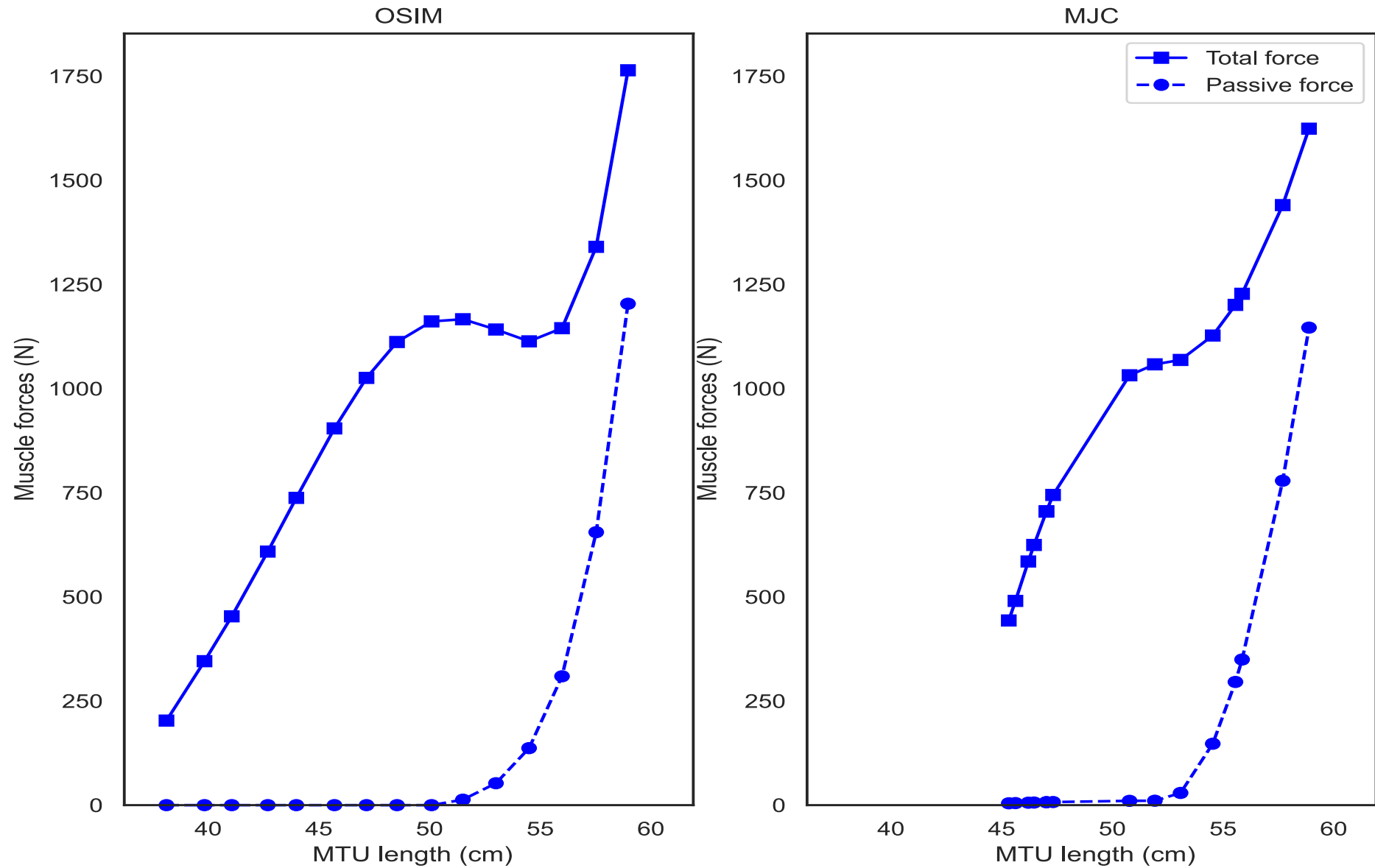
Muscle specific force comparison before/after optimization

psoas_r



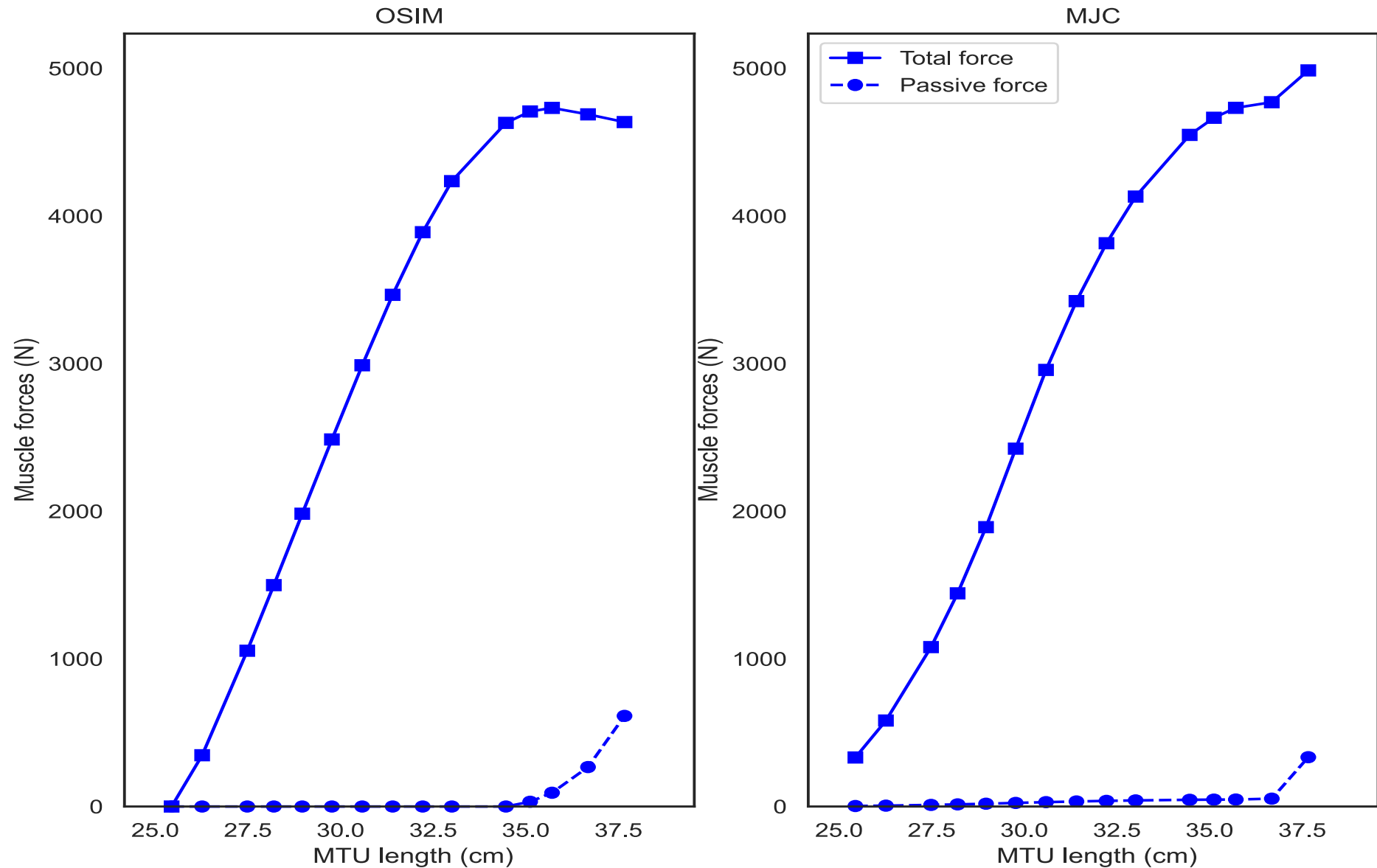
Muscle specific force comparison before/after optimization

rect_fem_r



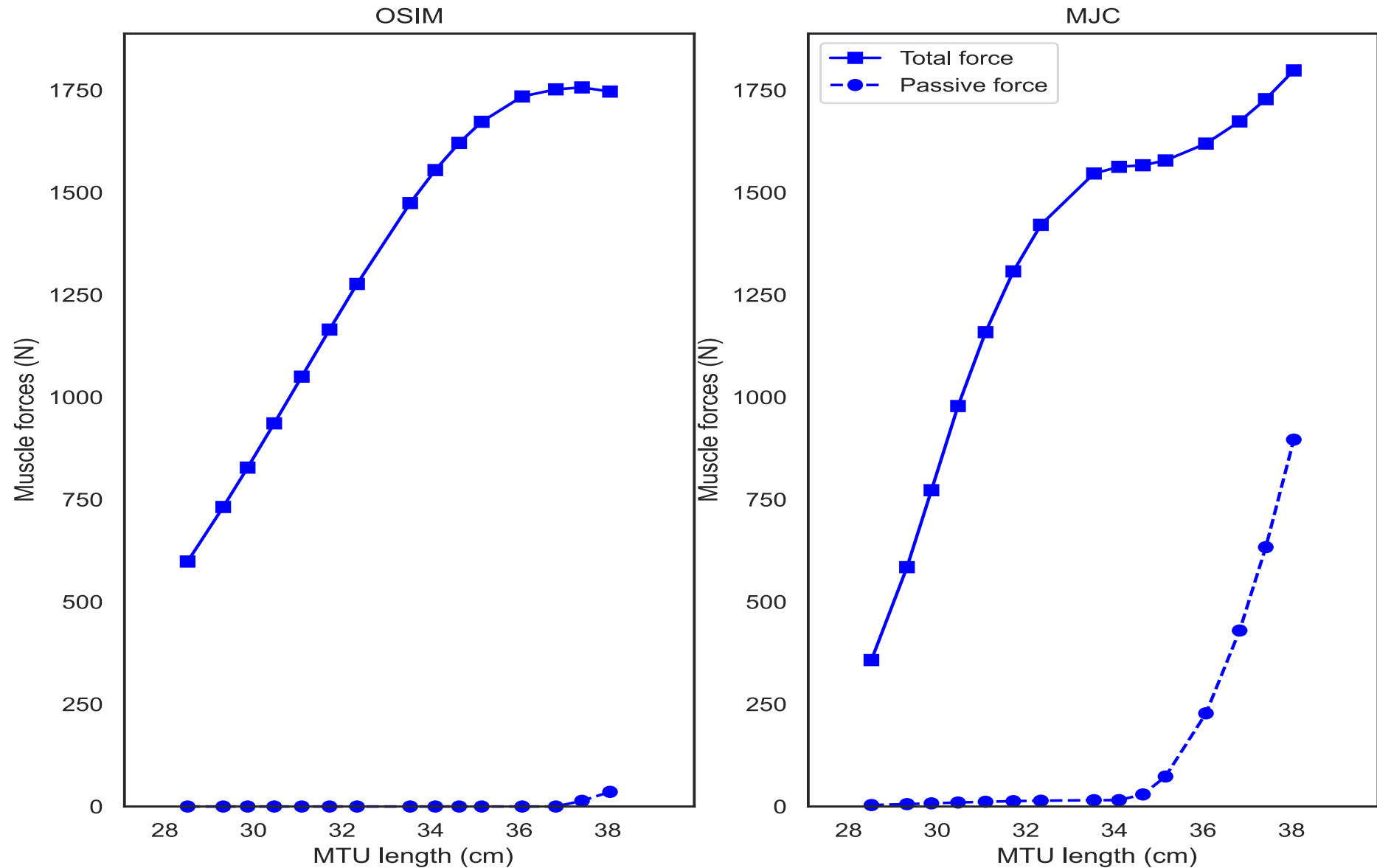
Muscle specific force comparison before/after optimization

soleus_r



Muscle specific force comparison before/after optimization

tib_ant_r



Muscle specific force comparison before/after optimization

vas_int_r

